
Iterated Negative Complex Roots

Principal-Branch Dynamics, Symbolic Itineraries, Exact Inverse
Trees,
Geometric Limits, Weighted Branch Measures, and
Infinite-Depth Structure

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Abstract. We develop an exact and parameterized theory of iterated complex roots under explicit branch conventions. The principal negative-root recursion

$$z_{n+1} = F_r(z_n) := \text{Root}_r^{\text{Pr}}(-z_n), \quad r \geq 2,$$

separates completely into radial and angular dynamics. The radius obeys the exact law $|z_n| = |z_0|^{1/r^n}$, while the principal argument is governed by a two-branch contraction. This yields a closed formula for the absolute angular variable, a global nonzero attracting two-cycle, a complete classification of periodic points, and sharp convergence estimates. We then enlarge the model to the phase-rotated family $F_{r,\beta}(z) = \text{Root}_r^{\text{Pr}}(e^{i\beta}z)$ and prove an exact periodic-itinerary principle: every admissible finite branch word determines at most one periodic orbit, with explicit angle formula and multiplier. This broader family exhibits genuine higher-period behavior, including explicit period-three examples for $r = 2$, showing that the negative-root case is a distinguished symmetric member rather than a generic periodic regime. On the multivalued side we identify recursive root generation with inverse iteration of the monomial $-z^r$, obtain exact regular-polygon formulas, exact Hausdorff distance to the unit circle, and equidistribution of normalized branch measures. We further introduce weighted branch measures, derive their limiting Bernoulli measures on the circle, and connect their entropy to information growth in the branch tree. Variable root orders are treated exactly at the algebraic level and quantitatively at the principal-dynamical level. Finally, we formalize the distinction among fixed-point, pointwise, set-valued, and measure-valued meanings of an infinite radical. The presentation is self-contained, but deliberately places the results in the classical literature on complex iterated radicals and the broader framework of complex dynamics and fractal measures.

Contents

1	Introduction	5
2	Complex roots, principal arguments, and branch conventions	6
2.1	Principal argument and logarithm	7
2.2	All roots of a nonzero complex number	7
2.3	A necessary distinction: root order versus recursion depth	7
3	The principal negative-root map	8
4	Exact radial dynamics	9
5	Exact angular dynamics	9
5.1	Sharp global convergence estimate	12
6	The unwrapped lift and the apparent fixed point	12
7	A phase-parameter family and symbolic periodicity	13
7.1	Normalization as a contracted rotation	13
7.2	The fixed-point regime	14
7.3	The finite symbolic alphabet	15
7.4	Periodic itineraries and exact candidate points	15
7.5	Periodic parameter windows	16
7.6	Explicit low-period windows for $r = 2$	16
7.7	Boundary collisions and local persistence	17
7.8	The negative-root case revisited	17
7.9	A computational theorem for window discovery	18
8	The unbranched and fully branched viewpoints	18
8.1	Exact monomial iteration	19
8.2	Branch histories and exact base- r coding	19
9	Geometry of the complete branch sets	20
9.1	Exact Hausdorff distance	20
9.2	Individual orbit asymptotics	21
9.3	Convex hulls, areas, and edge scales	22
10	Normalized branch measures and weighted branch measures	22
10.1	Uniform branch selection	22
10.2	Weighted branch probabilities and chronological reversal	24
10.3	The self-similar structure of the limit measure	25
10.4	Multifractal frequency classes	26

10.5	Uniformity, singularity, and the geometry of bias	27
11	A detailed square-root case study	27
11.1	The orbit beginning at $z_0 = i$	27
11.2	Changing the initial modulus	27
12	Variable root orders and nonautonomous contraction	28
12.1	Exact radial law	29
12.2	Exact principal absolute-angle recurrence	29
12.3	Periodic order sequences	29
12.4	Mixed-order multivalued algebra	30
12.5	A stronger nonautonomous interpretation	30
13	Infinite radicals: four non-equivalent meanings	31
13.1	Fixed-point meaning	31
13.2	Pointwise iterative meaning	31
13.3	Set-valued meaning	32
13.4	Measure-valued meaning	32
13.5	Order of limits	32
14	Branch-cut hitting and exceptional initial data	32
15	Relations with classical complex dynamics	32
16	What is special about the monomial model?	33
17	Further structural directions and open problems	34
17.1	Complete phase-parameter classification	34
17.2	Higher-period phase windows	34
17.3	Weighted multifractal limits	34
17.4	Perturbed maps	34
17.5	Branch collisions	35
17.6	Nonautonomous principal dynamics	35
17.7	Order of limits	35
17.8	Other metrics	35
18	Contracted-rotation normal form and arithmetic phase classes	35
18.1	The circle map and a canonical lift	35
18.2	Recovering the fixed-point regime from the normal form	36
18.3	Periodic orbits and rotation number	37
18.4	Arithmetic phase parameters	37
18.5	The irrational regime as a conditional alternative	38
18.6	A phase diagram in the $r = 2$ case	38

18.7 The symbolic and rotational descriptions are complementary	38
18.8 A precise boundary statement	39
18.9 Implication for future classification	39
19 Further structural consequences and scope	40
19.1 Why the monomial case is exactly solvable	40
19.2 Piecewise-contraction and contracted-rotation context	40
19.3 Parameter-space problems	41
19.4 Entropy and nonperiodic regimes	41
19.5 A hierarchy of limiting viewpoints	42
19.6 Originality and relationship with earlier complex-radical work	42
19.7 Methodological lessons	42
20 Common logical pitfalls and methodological lessons	43
20.1 A complex square root is not globally single-valued	43
20.2 The principal logarithm and the multivalued logarithm answer different questions	43
20.3 Argument reduction must occur at every step	43
20.4 Density is not pointwise convergence	43
20.5 Recursive labels do not create a new field	43
21 Worked exact examples and dependence on the root order	43
21.1 The orders $r = 2, 3, 4, 5$	43
21.2 A general initial-angle family	44
21.3 The square-root orbit from $z_0 = i$	44
21.4 Large-order behavior of the principal cycle	45
21.5 Large-order behavior of the phase family	45
21.6 An order-uniform synchronization estimate	46
21.7 Why the contraction does not imply point convergence	46
22 Synthesis of the main results	46
23 Conclusion	47
A Appendix A: explicit formulas for signed angles	48
B Appendix B: exact Hausdorff calculation	48
C Appendix C: Fourier proof of equidistribution	49
D Appendix D: proof of the variable-order formula	49
E Appendix E: principal phase family and branch words	49
F Appendix F: bibliographic and notational notes	50

F.1	Principal branch convention	50
F.2	Terminology	50
G	Appendix G: Detailed phase-window calculations	50
G.1	Affine dependence on the phase parameter	50
G.2	The admissibility inequalities as an interval intersection	51
G.3	Cyclic representatives and the meaning of a word	51
G.4	The period-three calculation in full	51
G.5	Why strict admissibility gives local persistence	52
G.6	A rationality consequence	52
G.7	Algebraic parameters and the broader arithmetic picture	53
H	Appendix H: Detailed branch coding and self-similar measures	53
H.1	Exact endpoint coding	53
H.2	The reversal phenomenon in probability	53
H.3	Cylinder masses	54
H.4	Proof of exact dimensionality	54
H.5	Frequency classes and combinatorial dimension	54
H.6	Uniform branch measure and Fourier cancellation	55
I	Appendix I: Detailed asymptotic expansions	55
I.1	Expansion of the radial factor	55
I.2	Expansion of the angular Hausdorff term	55
I.3	Dominance of the angular side	56
I.4	Individual orbit expansion	56
I.5	A special unit-modulus simplification	56
J	Appendix J: Detailed nonautonomous order analysis	57
J.1	Closed formula by repeated substitution	57
J.2	Unconditional forgetting	57
J.3	Periodic schedules	57
J.4	Mixed-order branch cardinalities	57
J.5	Periodic forcing and signed periods	58
J.6	Nonperiodic order schedules	58

1 Introduction

The operation of extracting a complex root is elementary, but it becomes structurally interesting as soon as one iterates it. The reason is not that the complex field ceases to be sufficient: it does not. Rather, complex roots are intrinsically multivalued, and a deterministic iteration requires a branch convention. Once a branch is fixed, the chosen determination can have discontinuities, and those discontinuities may become part of the dynamics instead of disappearing as a notational nuisance.

The model considered throughout most of this paper is

$$z_{n+1} = F_r(z_n), \quad F_r(z) = \text{Root}_r^{\text{pr}}(-z), \quad r \in \{2, 3, 4, \dots\}. \quad (1)$$

The square-root case,

$$z_{n+1} = \sqrt{-z_n},$$

is the motivating example. Starting from $z_0 = i$, one obtains

$$i, \quad \sqrt{-i}, \quad \sqrt{-\sqrt{-i}}, \quad \sqrt{-\sqrt{-\sqrt{-i}}}, \dots$$

A naive calculation on an unwrapped argument variable suggests an affine fixed point; the actual principal-branch process instead alternates between the upper and lower half-planes. The first purpose of this paper is to make that distinction exact.

There is an extensive classical literature on complex iterated radicals. Ogilvy posed a research problem on the limiting behavior of complex iterated radicals in the *American Mathematical Monthly* in 1970; Gerber and Rohde subsequently obtained substantial partial results, and Walker later treated an important fixed-branch iteration more systematically.¹ Jones provides a broader discussion of continued powers and roots, while Schuske and Thron treat periodic infinite radicals from a different, classical viewpoint. Moreno and García-Caballero connect nested complex square roots to Chebyshev polynomials.² An earlier short preprint by the present author established the principal negative-root two-cycle and the branch-cut mechanism in the special fixed-order setting [1]. The present manuscript substantially enlarges that preliminary treatment by developing the phase family, exact symbolic and inverse-tree descriptions, branch measures, asymptotic geometry, and variable-order extensions in a unified framework. This history is stated explicitly so that the scope of the present contribution is not confused with the earlier preliminary result.

The present paper is therefore deliberately conservative about originality. The goal is not to introduce a new number system and not to rebrand the subject of iterated complex radicals. Instead, we isolate a transparent family in which several distinct structures can be solved exactly and then place those structures into a common dynamical language. The resulting picture has five interacting layers:

- (i) a deterministic principal-branch orbit in $\mathbb{C}$;

¹Walker explicitly describes the convergence question as having been posed by Ogilvy and notes the partial solutions of Rohde and Gerber. This historical point is worth keeping visible because the present paper is not intended to claim that complex iterated radicals themselves are a new subject. See Ogilvy [2], Gerber [3], Rohde [4], and Walker [5].

²The Chebyshev-polynomial perspective of Moreno and García-Caballero is particularly useful as a reminder that explicit formulas for nested complex radicals can arise from algebraic identities quite different from the branch-cut dynamics used here [9].

- (ii) a real one-dimensional angular cocycle obtained from polar coordinates;
- (iii) a symbolic description of branch itineraries in a phase-parameter family;
- (iv) a complete inverse tree for the multivalued recursion;
- (v) geometric and measure-theoretic limits of the finite branch sets.

The central fixed-order theorem is surprisingly rigid. For the negative-root map, every nonzero orbit has radius tending to 1 and argument magnitude tending to $\pi/(r+1)$. Hence the full orbit does not converge to a point: it approaches the two-cycle

$$\mathcal{C}_r := \{e^{i\pi/(r+1)}, e^{-i\pi/(r+1)}\}. \quad (2)$$

At the same time, if all root branches are retained, the depth- n set has r^n points, is a regular r^n -gon, and tends to $\mathbb{S}^1$ in the Hausdorff metric. The deterministic and multivalued limits are thus genuinely different objects.

The main new structural extension in this version is the phase family

$$F_{r,\beta}(z) := \text{Root}_r^{\text{pr}}(e^{i\beta}z), \quad \beta \in \mathbb{R}. \quad (3)$$

The negative-root system corresponds to $\beta = \pi$. The associated angular map is a piecewise-affine circle contraction with an integer-valued branch index. This allows periodic orbits to be encoded by finite words and, unlike the symmetric case $\beta = \pi$, permits higher periods. For instance, for $r = 2$ and $\beta = 2\pi/3$ there is an explicit period-three angular orbit.

A second extension concerns the branch tree. The uniform counting measure on the regular polygons converges to Haar measure on the circle, but uniformity is not mathematically forced. If each branch digit is chosen according to a probability vector $p = (p_0, \dots, p_{r-1})$, the branch tree carries a Bernoulli measure. The limiting endpoint measure is then a pushforward of the random base- r expansion

$$X = \sum_{j=1}^{\infty} \frac{K_j}{r^j}, \quad \mathbb{P}(K_j = k) = p_k,$$

and is generally non-Haar. Thus the recursive root process naturally links complex dynamics with symbolic probability and entropy.

The organization is therefore as follows. Sections 2–4 establish the branch conventions and solve the principal negative-root dynamics exactly. Section 5 explains the unwrapped lift and why its fixed point is not the principal-branch answer. Section 6 develops the phase-parameter family and the periodic-itinerary theorem. Sections 7–9 analyze the full multivalued recursion, its polygonal geometry, and weighted branch measures. Section 10 treats variable root orders. Section 11 revisits the infinite radical question using four distinct notions of convergence. Section 12 records further structural consequences, and Section 13 discusses the literature and the precise scope of the contribution. Several appendices collect proofs and asymptotic expansions.

Central message. A recursive complex root is not a mysterious new arithmetic operation. What becomes nontrivial is the interaction between a branch convention, iteration, and the information carried by branch histories. In the present model, that interaction is explicit enough to admit exact theorems rather than heuristic numerical claims.

2 Complex roots, principal arguments, and branch conventions

2.1 Principal argument and logarithm

We use the principal argument convention

$$\operatorname{Arg} z \in (-\pi, \pi], \quad z \neq 0. \quad (4)$$

Thus every nonzero $z \in \mathbb{C}$ has the representation

$$z = |z|e^{i \operatorname{Arg} z}. \quad (5)$$

The associated principal logarithm is

$$\operatorname{Log} z = \ln |z| + i \operatorname{Arg} z. \quad (6)$$

With this convention, the principal r th root is

$$\operatorname{Root}_r^{\text{Pr}}(z) := \exp\left(\frac{1}{r} \operatorname{Log} z\right) = |z|^{1/r} e^{i \operatorname{Arg}(z)/r}. \quad (7)$$

This is the standard principal-branch construction associated with the principal logarithm.³

A branch point occurs at 0, while the standard branch cut for the logarithm and principal power is the nonpositive real axis in the argument variable. Composing with multiplication by -1 moves the discontinuity to the nonnegative real axis in the original z -plane. This simple geometric shift is the source of the branch-induced sign alternation studied below.

2.2 All roots of a nonzero complex number

If

$$w = \rho e^{i\phi}, \quad \rho > 0,$$

then the complete set of r th roots of w is

$$\mathcal{R}_r(w) = \left\{ \rho^{1/r} \exp\left(\frac{i(\phi + 2\pi k)}{r}\right) : k = 0, 1, \dots, r-1 \right\}. \quad (8)$$

All r roots have modulus $\rho^{1/r}$ and their arguments are equally spaced by $2\pi/r$.

Proposition 2.1 (Regular-polygon form). *For $w \neq 0$, the set $\mathcal{R}_r(w)$ is the vertex set of a regular r -gon centered at the origin. For $w = 0$, the root set is $\{0\}$.*

Proof. Equation (8) gives a common modulus and equally spaced arguments. Distinctness follows because the differences in the branch indices are strictly smaller than r . $\square$

2.3 A necessary distinction: root order versus recursion depth

The principal n th root of i is

$$\operatorname{Root}_n^{\text{Pr}}(i) = \exp\left(\frac{i\pi}{2n}\right) \longrightarrow 1. \quad (9)$$

³The NIST Digital Library of Mathematical Functions gives the principal logarithm as the single-valued analytic branch on $\mathbb{C} \setminus (-\infty, 0]$, with the principal argument convention used here. See DLMF §4.2 [8].

This is not the same operation as repeatedly applying the principal square root after negation. In one problem the exponent changes while the input remains fixed; in the other the output is recursively fed back into the map. Likewise, taking all n th roots of a fixed number produces a regular n -gon, while taking all branches through n recursive stages of a fixed root produces a regular r^n -gon. Confusing these constructions is a common source of misleading “infinite root” arguments.

3 The principal negative-root map

Definition 3.1 (Principal negative r th-root operator). *For an integer $r \geq 2$, define*

$$F_r : \mathbb{C} \rightarrow \mathbb{C}, \quad F_r(z) = \text{Root}_r^{\text{pr}}(-z). \quad (10)$$

The associated principal negative-root system is

$$z_{n+1} = F_r(z_n).$$

We first make the branch reduction exact.

Lemma 3.2 (Polar form). *Let $z = \rho e^{i\theta}$ with $\rho > 0$ and $\theta \in (-\pi, \pi]$. Then*

$$F_r(z) = \rho^{1/r} e^{iT_r(\theta)}, \quad (11)$$

where

$$T_r(\theta) = \begin{cases} \frac{\theta + \pi}{r}, & -\pi < \theta \leq 0, \\ \frac{\theta - \pi}{r}, & 0 < \theta \leq \pi. \end{cases} \quad (12)$$

Moreover, $F_r(0) = 0$.

Proof. If $-\pi < \theta \leq 0$, then $\theta + \pi \in (0, \pi]$, so it is already the principal argument of $-z$. If $0 < \theta \leq \pi$, then $\theta + \pi \in (\pi, 2\pi]$, and its principal representative is $\theta - \pi \in (-\pi, 0]$. Applying (7) proves the formula. The value at 0 is defined by $F_r(0) = 0$. $\square$

Proposition 3.3 (Image wedge and discontinuity). *For every $\theta \in (-\pi, \pi]$,*

$$T_r(\theta) \in \left(-\frac{\pi}{r}, \frac{\pi}{r}\right]. \quad (13)$$

Consequently, after one iteration every nonzero orbit lies in the wedge

$$W_r := \left\{ \rho e^{i\theta} : \rho > 0, |\theta| \leq \frac{\pi}{r} \right\}. \quad (14)$$

The map F_r is continuous on $\mathbb{C} \setminus [0, \infty)$ and at 0, but discontinuous at every $x > 0$.

Proof. The range statement follows directly from the two affine branches. If $x > 0$, the upper and lower approaches to x induce opposite one-sided approaches to the negative real axis in the argument of $-z$ and hence give limiting values

$$x^{1/r} e^{-i\pi/r} \quad \text{and} \quad x^{1/r} e^{i\pi/r},$$

which differ. Continuity at 0 follows from $|F_r(z)| = |z|^{1/r} \rightarrow 0$. $\square$

Remark 3.4. *The branch discontinuity should be regarded as part of the dynamical law. The correct angular recurrence is not the unrestricted affine relation $(\theta + \pi)/r$; the principal reduction must be applied at every step. This is precisely the distinction that later produces the two-cycle.*

4 Exact radial dynamics

Let

$$z_n = \rho_n e^{i\theta_n}, \quad \rho_n > 0, \quad \theta_n \in (-\pi, \pi]. \quad (15)$$

Proposition 4.1 (Exact radial law). *For every nonzero initial point z_0 ,*

$$\rho_{n+1} = \rho_n^{1/r}, \quad \boxed{\rho_n = \rho_0^{1/r^n}}. \quad (16)$$

Consequently,

$$\rho_n \rightarrow 1. \quad (17)$$

Proof. The modulus of an r th root is the r th root of the modulus. Iteration gives the exact formula. Equivalently,

$$\log \rho_n = r^{-n} \log \rho_0 \rightarrow 0.$$

Exponentiation yields $\rho_n \rightarrow 1$. □

Corollary 4.2 (Exact radial error). *For every $n \geq 0$,*

$$|\log \rho_n| = r^{-n} |\log \rho_0|. \quad (18)$$

Moreover,

$$|\rho_n - 1| \leq \frac{|\log \rho_0|}{r^n} \exp\left(\frac{|\log \rho_0|}{r^n}\right). \quad (19)$$

Proof. The first formula is immediate. The second follows from $|e^u - 1| \leq |u|e^{|u|}$ with $u = \log \rho_n$. □

Remark 4.3. *There is no branch competition in the radial coordinate: all branches generated from the same starting point have the same modulus at a fixed depth. All interesting distinctions between branch selections therefore live in the angular coordinate.*

5 Exact angular dynamics

Define

$$\alpha_n := |\theta_n| \in [0, \pi]. \quad (20)$$

From (12), the angular map has an exact absolute-value reduction.

Proposition 5.1 (Sign reversal and scalar contraction). *For every $\theta \neq 0$,*

$$\operatorname{sgn} T_r(\theta) = -\operatorname{sgn}(\theta), \quad (21)$$

and for every $\theta \in (-\pi, \pi]$,

$$|T_r(\theta)| = \frac{\pi - |\theta|}{r}. \quad (22)$$

Therefore

$$\alpha_{n+1} = G_r(\alpha_n), \quad G_r(x) = \frac{\pi - x}{r}. \quad (23)$$

Proof. If $\theta > 0$ then $(\theta - \pi)/r < 0$; if $\theta < 0$ then $(\theta + \pi)/r > 0$. Taking absolute values gives (22); the formula also holds for $\theta = 0$. $\square$

The scalar map G_r is an orientation-reversing contraction with Lipschitz constant $1/r$. Its fixed point is

$$\alpha_* := \frac{\pi}{r+1}. \quad (24)$$

Theorem 5.2 (Exact absolute-angle formula). *For every principal orbit,*

$$\boxed{\alpha_n = \frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right)}. \quad (25)$$

In particular,

$$|\alpha_n - \alpha_*| = r^{-n} |\alpha_0 - \alpha_*|, \quad (26)$$

and hence $\alpha_n \rightarrow \pi/(r+1)$.

Proof. The fixed-point equation $x = (\pi - x)/r$ gives $x = \pi/(r+1)$. Subtracting the fixed-point identity from (23) yields

$$\alpha_{n+1} - \alpha_* = -\frac{1}{r}(\alpha_n - \alpha_*).$$

Iteration gives the result. $\square$

Corollary 5.3 (Signed asymptotic alternation). *Except for the single transient orbit beginning at $\theta_0 = \pi$, the signs of θ_n alternate from the first step onward. Consequently the even and odd subsequences approach opposite points of the limiting two-cycle.*

Proof. The sign-reversal proposition gives alternation whenever $\theta_n \neq 0$. The unique way to obtain $\theta_{n+1} = 0$ is $\theta_n = \pi$. For that case, $\theta_1 = 0$ and $\theta_2 = \pi/r > 0$, after which alternation resumes. $\square$

Proposition 5.4 (Two-step map). *For $\theta \neq 0$,*

$$T_r^2(\theta) = \begin{cases} \frac{\theta - \pi(r-1)}{r^2}, & \theta < 0, \\ \frac{\theta + \pi(r-1)}{r^2}, & \theta > 0. \end{cases} \quad (27)$$

Thus the second iterate is affine with slope r^{-2} on each sign sector.

Proof. If $\theta < 0$, then $T_r(\theta) > 0$ and the second application uses the positive branch:

$$T_r^2(\theta) = \frac{1}{r} \left(\frac{\theta + \pi}{r} - \pi \right) = \frac{\theta - \pi(r-1)}{r^2}.$$

The other case is symmetric. $\square$

Theorem 5.5 (Unique attracting two-cycle). *For every integer $r \geq 2$, the points*

$$a_r := e^{i\pi/(r+1)}, \quad b_r := e^{-i\pi/(r+1)} \quad (28)$$

form a period-two orbit:

$$F_r(a_r) = b_r, \quad F_r(b_r) = a_r. \quad (29)$$

Moreover, every nonzero orbit satisfies

$$\text{dist}(z_n, \{a_r, b_r\}) \rightarrow 0. \quad (30)$$

Proof. The angular identities are

$$T_r\left(\frac{\pi}{r+1}\right) = -\frac{\pi}{r+1}, \quad T_r\left(-\frac{\pi}{r+1}\right) = \frac{\pi}{r+1},$$

and the cycle points have modulus one. For a general nonzero orbit, the radius tends to one and the argument magnitude tends to $\pi/(r+1)$, while the sign alternates. The Euclidean estimate

$$|\rho e^{i\theta} - e^{i\varphi}| \leq |\rho - 1| + |\theta - \varphi|$$

then implies convergence to the corresponding cycle point. $\square$

Theorem 5.6 (Complete periodic-point classification). *For every $r \geq 2$, the map F_r has exactly one fixed point, namely 0, and exactly one nonzero periodic orbit, namely*

$$\{a_r, b_r\}.$$

There are no nonzero periodic orbits of period greater than two.

Proof. Suppose $z_0 \neq 0$ lies on a periodic orbit of period p . The radial identity gives

$$\rho_0 = \rho_0^{1/r^p},$$

so $\rho_0 = 1$. The angular magnitude formula gives

$$\alpha_p - \alpha_* = \left(-\frac{1}{r}\right)^p (\alpha_0 - \alpha_*).$$

Periodicity implies $\alpha_p = \alpha_0$, hence $\alpha_0 = \alpha_*$. Thus every point on the periodic orbit has argument magnitude $\pi/(r+1)$. Since the sign reverses at every nonzero step, the orbit consists of exactly the two points a_r, b_r . For a fixed point, sign reversal rules out any nonzero argument, while angle zero maps to $\pi/r \neq 0$; therefore only 0 remains. $\square$

Proposition 5.7 (Local multiplier of the cycle). *On any branch chart avoiding the branch cut,*

$$F'_r(z) = \frac{F_r(z)}{rz}. \quad (31)$$

At the two-cycle,

$$F'_r(a_r)F'_r(b_r) = \frac{1}{r^2}. \quad (32)$$

Thus the cycle is locally holomorphically attracting with multiplier r^{-2} .

Proof. Differentiate $\exp((1/r)\text{Log}(-z))$ on a branch chart. At a_r, b_r the product telescopes to $1/r^2$. $\square$

Remark 5.8. We use the phrase “locally holomorphically attracting” rather than asserting that F_r is a globally holomorphic dynamical system: the principal map has a genuine discontinuity on $(0, \infty)$. The cycle itself lies away from that discontinuity, so local multiplier theory is entirely standard there.

5.1 Sharp global convergence estimate

Let $\varphi_n \in \{\pm\pi/(r+1)\}$ be the cycle angle with the same sign as θ_n once the transient zero-angle case has passed. Then

$$|\theta_n - \varphi_n| = \frac{1}{r^n} \left| \alpha_0 - \frac{\pi}{r+1} \right|. \quad (33)$$

Combining this with the radial error gives

$$\text{dist}(z_n, \mathcal{C}_r) \leq \frac{|\log \rho_0|}{r^n} \exp\left(\frac{|\log \rho_0|}{r^n}\right) + \frac{1}{r^n} \left| \alpha_0 - \frac{\pi}{r+1} \right|. \quad (34)$$

In particular,

$$\text{dist}(z_n, \mathcal{C}_r) = O(r^{-n}). \quad (35)$$

6 The unwrapped lift and the apparent fixed point

A common calculation starts from the angular recurrence

$$\tilde{T}_r(\theta) = \frac{\theta + \pi}{r} \quad (36)$$

and then iterates it on the full real line. This is a legitimate dynamical system, but it is not the principal-argument dynamics.

Proposition 6.1 (Exact unwrapped solution). *For every $\theta_0 \in \mathbb{R}$,*

$$\tilde{T}_r^n(\theta_0) = \frac{\theta_0}{r^n} + \frac{\pi}{r-1}(1 - r^{-n}). \quad (37)$$

Consequently,

$$\tilde{T}_r^n(\theta_0) \rightarrow \frac{\pi}{r-1}. \quad (38)$$

Proof. This is the standard solution of an affine recurrence. $\square$

For $r = 2$, the unwrapped fixed point is π , whose exponential is -1 . It is tempting to declare -1 to be the value of the infinite radical. That conclusion is not correct for the principal recursion, because the argument variable must be reduced after every multiplication by -1 .

Indeed,

$$T_r(\theta) = \begin{cases} \tilde{T}_r(\theta), & \theta \leq 0, \\ \tilde{T}_r(\theta) - \frac{2\pi}{r}, & \theta > 0. \end{cases} \quad (39)$$

The correction term is discontinuous in sign, and the correction is precisely what changes a putative fixed point into a two-cycle.

Remark 6.2. The unwrapped fixed point is not “wrong” as a statement about the lifted recurrence. The error arises only if one identifies the lifted recurrence with the principal complex iteration.

Iteration and reduction modulo 2π do not commute.

7 A phase-parameter family and symbolic periodicity

The negative sign is a distinguished phase, but it is natural to ask what survives when the factor $-1 = e^{i\pi}$ is replaced by a general unimodular phase. This enlarges the problem from a single rigid orbit structure to a one-parameter family of discontinuous contractions.

Definition 7.1 (Phase-rotated principal root map). *For $r \geq 2$ and $\beta \in \mathbb{R}$, define*

$$F_{r,\beta}(z) := \text{Root}_r^{\text{pr}}(e^{i\beta}z). \quad (40)$$

Since only $e^{i\beta}$ enters, the parameter is 2π -periodic and we shall usually work with $\beta \in (-\pi, \pi]$.

Definition 7.2 (Principal-value reduction). *For $x \in \mathbb{R}$, let $\text{PV}(x)$ be the unique representative of x modulo 2π lying in $(-\pi, \pi]$. Thus*

$$\text{PV}(x) = x - 2\pi k, \quad -\pi < x - 2\pi k \leq \pi, \quad (41)$$

for a unique integer k .

If $z = \rho e^{i\theta}$ with $\theta \in (-\pi, \pi]$, then

$$F_{r,\beta}(z) = \rho^{1/r} e^{iT_{r,\beta}(\theta)}, \quad T_{r,\beta}(\theta) = \frac{\text{PV}(\theta + \beta)}{r}. \quad (42)$$

Thus the radial coordinate remains completely explicit and all new dynamical structure is angular.

7.1 Normalization as a contracted rotation

After one iteration, every angle belongs to

$$I_r := \left(-\frac{\pi}{r}, \frac{\pi}{r} \right]. \quad (43)$$

Introduce the normalized coordinate

$$x = \frac{r\theta + \pi}{2\pi}. \quad (44)$$

Then $\theta = (2\pi x - \pi)/r$, and a direct substitution into (42) gives

$$\boxed{x_{n+1} = \left\{ \frac{x_n}{r} + \delta_\beta \right\}}, \quad (45)$$

where

$$\delta_\beta = \frac{r-1}{2r} + \frac{\beta}{2\pi}. \quad (46)$$

Here $\{\cdot\}$ denotes the fractional part, with the half-open endpoint convention inherited from $(-\pi, \pi]$. This identity is exact, not an analogy: the angular dynamics of the phase family are a concrete circle contraction with slope $1/r$.

The normalized form places the phase family in the classical setting of contracted rotations $x \mapsto \{\lambda x + \delta\}$, $0 < \lambda < 1$. Laurent and Nogueira study this family, including its rotation number

and the symbolic structure generated by the discontinuity [17]. Their work shows, in particular, that an irrational rotation number leads to a Cantor limit set carrying dynamics conjugate to an irrational circle rotation, while a rational rotation number yields at most one periodic orbit in the standard contracted-rotation setting [17]. These facts provide useful context for the present family, but they should not be confused with a classification theorem proved here for the special affine subfamily induced by the root map.

More broadly, the phase family belongs to the theory of piecewise contractions. Recent work of Jain and Liverani establishes generic periodic behavior and topological stability for broad classes of piecewise injective contractions, while also emphasizing the exceptional role of discontinuity-attractor collisions [20]. The present system is substantially more explicit: every affine piece has exactly the same slope $1/r$, the discontinuity is known in closed form, and periodic candidates can therefore be computed algebraically before admissibility is checked.

7.2 The fixed-point regime

The first regime is the one in which the principal reduction disappears after the initial image step.

Theorem 7.3 (Necessary and sufficient fixed-point regime). *Let $r \geq 2$ and $\beta \in (-\pi, \pi]$. The phase map $T_{r,\beta}$ has a fixed point if and only if*

$$|\beta| \leq \pi - \frac{\pi}{r}. \quad (47)$$

In that case the fixed point is unique and is given by

$$\theta_*(\beta) = \frac{\beta}{r-1}. \quad (48)$$

Moreover, every principal angular orbit enters the no-further-reduction affine regime after at most one iterate and then converges geometrically to $\theta_(\beta)$.*

Proof. If (47) holds, then for every $\theta_1 \in I_r$ one has $-\pi < \theta_1 + \beta \leq \pi$, so

$$T_{r,\beta}(\theta_1) = \frac{\theta_1 + \beta}{r}.$$

The latter affine contraction has unique fixed point $\beta/(r-1)$ and ratio $1/r$. The bound on β is precisely the condition that this fixed point belongs to I_r .

Conversely, suppose θ is a fixed point. For some $k \in \mathbb{Z}$,

$$r\theta = \theta + \beta - 2\pi k, \quad \theta = \frac{\beta - 2\pi k}{r-1}.$$

Since $\theta \in I_r$, we have $|\theta| \leq \pi/r$. If $k \neq 0$, then $|\beta - 2\pi k| \geq \pi$, and therefore $|\theta| \geq \pi/(r-1) > \pi/r$, a contradiction. Thus $k = 0$, so $|\beta| \leq (r-1)\pi/r = \pi - \pi/r$. $\square$

For $\beta = 0$ this recovers the familiar convergence of the principal r th-root iteration to 1. The negative-root value $\beta = \pi$ is exactly on the opposite side of the fixed-point threshold and is therefore forced into branch-switching dynamics.

7.3 The finite symbolic alphabet

For $\beta \in (-\pi, \pi]$ and a principal angle $\theta \in (-\pi, \pi]$, one has

$$-2\pi < \theta + \beta \leq 2\pi.$$

Consequently the principal-reduction index satisfies

$$k \in \{-1, 0, 1\}. \quad (49)$$

Thus, although it is convenient to write $k \in \mathbb{Z}$, every actual orbit in the normalized phase family uses the finite alphabet $\{-1, 0, 1\}$. This elementary observation is important because it turns the phase analysis into a finite-word problem at every fixed period.

At step j , the branch equation is

$$\theta_{j+1} = \frac{\theta_j + \beta - 2\pi k_j}{r}, \quad k_j \in \{-1, 0, 1\}, \quad (50)$$

and admissibility is exactly

$$-\pi < \theta_j + \beta - 2\pi k_j \leq \pi. \quad (51)$$

The strict version replaces the right inequality by $< \pi$ as well and therefore excludes all branch-boundary collisions.

7.4 Periodic itineraries and exact candidate points

Definition 7.4 (Admissible chronological word). *A word $\mathbf{k} = (k_0, \dots, k_{p-1})$ with letters in $\{-1, 0, 1\}$ is admissible for (r, β) if the affine recurrence (50) with those branch indices has a p -periodic solution satisfying (51) at each step. It is strictly admissible when all branch inequalities are strict.*

Theorem 7.5 (Periodic-itinerary formula). *Fix $r \geq 2$, $\beta \in (-\pi, \pi]$, and a chronological word $\mathbf{k} = (k_0, \dots, k_{p-1})$. Put*

$$S_p := 1 + r + \dots + r^{p-1} = \frac{r^p - 1}{r - 1}, \quad (52)$$

and define

$$K_p^{\text{chron}}(\mathbf{k}) := k_0 + k_1 r + \dots + k_{p-1} r^{p-1}. \quad (53)$$

Then any p -periodic orbit with this chronological itinerary must begin at

$$\boxed{\theta_0 = \frac{\beta S_p - 2\pi K_p^{\text{chron}}(\mathbf{k})}{r^p - 1}}. \quad (54)$$

Conversely, if the orbit generated from this angle satisfies the admissibility inequalities, then it is p -periodic. If the word is strictly admissible, its local p -step multiplier is r^{-p} .

Proof. Repeated substitution gives

$$\theta_p = \frac{\theta_0 + \beta S_p - 2\pi(k_0 + k_1 r + \dots + k_{p-1} r^{p-1})}{r^p}.$$

Solving $\theta_p = \theta_0$ gives (54). The converse is exactly the admissibility condition. On a fixed itinerary the derivative of every branch is $1/r$, so the derivative of the composition is r^{-p} . $\square$

A primitive word gives a candidate for least period p . More precisely, when the word is strictly admissible and primitive, the resulting orbit has least period p : a smaller period would force the branch itinerary itself to be periodic with a smaller period, contradicting primitivity.

Proposition 7.6 (Finite candidate bound). *For fixed r , β , and period p , there are at most 3^p chronological words and therefore at most 3^p candidate periodic points before identifications by cyclic shifts. The exact dynamical problem is thus finite at every prescribed period.*

Proof. There are three admissible branch symbols at each of the p steps. Each word determines at most one candidate angle by (54). $\square$

7.5 Periodic parameter windows

Write $b = \beta/\pi$. For a fixed word, the candidate angle $u_0(b) := \theta_0/\pi$ is affine in b , and every subsequent $u_j(b)$ is affine in b . Hence all admissibility inequalities are strict linear inequalities in one real parameter.

Theorem 7.7 (Periodic-window principle). *Fix $r \geq 2$ and a finite chronological word $\mathbf{k}$. The set of $\beta \in (-\pi, \pi]$ for which $\mathbf{k}$ is strictly admissible is either empty or an open interval relative to the interior of the chosen parameter domain. Its interior endpoints are obtained from branch-boundary equations*

$$u_j(b) + b - 2k_j = \pm 1. \quad (55)$$

Inside the interval, the entire periodic orbit depends affinely on β and its symbolic itinerary is locally constant.

Proof. The candidate $u_0(b)$ is affine. Inductively, $u_{j+1}(b) = (u_j(b) + b - 2k_j)/r$ is affine. Each strict admissibility condition is therefore an open interval in b (or the empty set) and the finite intersection of such sets is again an open interval or the empty set. An endpoint inside the parameter domain occurs when at least one inequality becomes an equality. Away from the resulting finite boundary set, all inequalities retain their signs under sufficiently small perturbations of b , so the itinerary persists. $\square$

The statement is intentionally phrased with “interior endpoints.” The global parameter domain $(-\pi, \pi]$ itself is half-open, and its endpoints are a separate source of one-sided behavior.

7.6 Explicit low-period windows for $r = 2$

Example 7.8 (Period three). *For $r = 2$ and the chronological word $(1, 0, 0)$,*

$$S_3 = 7, \quad K_3^{\text{chron}} = 1,$$

so

$$\theta_0 = \pi \left(b - \frac{2}{7} \right), \quad b = \frac{\beta}{\pi}.$$

The three strict branch inequalities reduce to

$$-1 < 2b - \frac{16}{7} < 1, \quad -1 < 2b - \frac{8}{7} < 1, \quad -1 < 2b - \frac{4}{7} < 1,$$

and their intersection is

$$\boxed{\frac{9}{14} < b < \frac{11}{14}}. \quad (56)$$

Thus

$$\frac{9\pi}{14} < \beta < \frac{11\pi}{14}$$

is a strict period-three window. At $\beta = 2\pi/3$ one obtains

$$\theta_0 = \frac{8\pi}{21}, \quad \theta_1 = -\frac{10\pi}{21}, \quad \theta_2 = \frac{2\pi}{21},$$

with multiplier $2^{-3} = 1/8$.

Example 7.9 (Periods four and five). *The primitive words $(0, 0, 0, 1)$ and $(0, 0, 0, 0, 1)$ yield, respectively,*

$$\frac{17\pi}{30} < \beta < \frac{19\pi}{30}, \tag{57}$$

and

$$\frac{33\pi}{62} < \beta < \frac{35\pi}{62}. \tag{58}$$

These exact rational endpoints arise from finite linear admissibility tests; no numerical root search is required.

The examples show that the phase parameter creates genuine higher-period structure, but they do not by themselves classify the entire parameter axis. Complete enumeration of the primitive windows for a fixed r remains a separate combinatorial problem.

7.7 Boundary collisions and local persistence

The principal convention $(-\pi, \pi]$ treats the two endpoints asymmetrically. A branch value equal to $-\pi$ is excluded, while a value equal to $+\pi$ is included. Therefore strict and non-strict admissibility should never be conflated.

For a strictly admissible word, all branch inequalities hold with a positive margin. By continuity of the affine candidate angles in β , there is then an open neighborhood of the parameter in which the same inequalities remain valid. Hence the periodic orbit persists, its coordinates remain affine functions of β , and the multiplier remains r^{-p} . The destruction of such persistence can occur only when the orbit meets a branch boundary.

This gives a concrete bifurcation mechanism: an exact periodic window is a connected parameter region on which a single symbolic itinerary survives, and the boundary is detected by an explicit equality before any numerical simulation is attempted.

7.8 The negative-root case revisited

For $\beta = \pi$ the phase family becomes the original negative-root system. Here the general symbolic complexity collapses under the absolute-angle transformation of equation (25). Every nonzero orbit is forced toward

$$\{e^{i\pi/(r+1)}, e^{-i\pi/(r+1)}\},$$

and no higher nonzero period can occur. The contrast is structural: the phase family admits symbolic branch switching, while the symmetric negative phase possesses an additional absolute-value reduction that removes that freedom entirely.

This is one of the central reasons to retain both viewpoints. The contracted-rotation normalization explains where the family sits in one-dimensional dynamics; the exact absolute-angle reduction explains why the negative-root member is unusually rigid.

7.9 A computational theorem for window discovery

The preceding discussion can be turned into an exact algorithm.

Proposition 7.10 (Finite symbolic window algorithm). *Fix $r \geq 2$ and $p \geq 1$. To find all strict period- p windows of the phase family on $(-\pi, \pi)$ it suffices to:*

- (1) *enumerate the 3^p words in $\{-1, 0, 1\}^p$;*
- (2) *discard nonprimitive words if least period p is desired;*
- (3) *compute the unique candidate angle from (54);*
- (4) *propagate the p affine angle formulas;*
- (5) *intersect the resulting $2p$ strict linear inequalities;*
- (6) *retain the nonempty intersections.*

The result is an exact finite collection of open intervals, together with their explicit affine periodic branches.

Proof. Each step is algebraic and finite. The only branch selection test is the pair of strict inequalities at each of the p steps, so no numerical dynamical approximation is needed. $\square$

This algorithmic form is useful even when one does not carry out the enumeration for large p . It identifies the correct object of study: not a numerical bifurcation diagram generated by sampling, but an exact arrangement of finitely many affine boundary functions.

8 The unbranched and fully branched viewpoints

The deterministic map keeps one principal branch. To retain all branches, define the correspondence

Definition 8.1 (Full negative-root correspondence). *For $z \in \mathbb{C}$, let*

$$\mathcal{F}_r(z) := \{w \in \mathbb{C} : w^r = -z\}. \quad (59)$$

For $z \neq 0$, the set has exactly r elements.

Introduce the polynomial

$$P_r(w) := -w^r. \quad (60)$$

Then

$$w \in \mathcal{F}_r(z) \iff P_r(w) = z. \quad (61)$$

Thus $\mathcal{F}_r$ is exactly the inverse correspondence of P_r .

Definition 8.2 (Depth- n branch set). *Starting from $A_0 = \{z_0\}$, define recursively*

$$A_{n+1} := \bigcup_{z \in A_n} \mathcal{F}_r(z). \quad (62)$$

The set A_n consists of all values obtained after n recursive root choices.

Proposition 8.3 (Inverse-iteration identity). *For every $n \geq 0$,*

$$A_n = P_r^{-n}(\{z_0\}) = \{w : P_r^{\circ n}(w) = z_0\}. \quad (63)$$

Proof. The claim is immediate for $n = 0$. If $A_n = P_r^{-n}(\{z_0\})$, then

$$A_{n+1} = P_r^{-1}(A_n) = P_r^{-(n+1)}(\{z_0\}).$$

□

8.1 Exact monomial iteration

Lemma 8.4. *Let*

$$S_n := 1 + r + r^2 + \cdots + r^{n-1}. \quad (64)$$

Then

$$P_r^{\circ n}(w) = (-1)^{S_n} w^{r^n}. \quad (65)$$

Proof. Induction gives

$$P_r^{\circ(n+1)}(w) = - \left((-1)^{S_n} w^{r^n} \right)^r = (-1)^{1+rS_n} w^{r^{n+1}},$$

and $1 + rS_n = S_{n+1}$. □

Theorem 8.5 (Closed depth equation). *For $z_0 \neq 0$,*

$$\boxed{A_n = \{w : w^{r^n} = (-1)^{S_n} z_0\}}. \quad (66)$$

Therefore

$$|A_n| = r^n. \quad (67)$$

Proof. By (63) and (65), $w \in A_n$ exactly when

$$(-1)^{S_n} w^{r^n} = z_0,$$

which is equivalent to (66). A nonzero complex number has exactly r^n distinct r^n th roots. □

Corollary 8.6 (Parity structure). *If r is even, then S_n is odd for every n , so*

$$A_n = \{w : w^{r^n} = -z_0\}. \quad (68)$$

If r is odd, then $S_n \equiv n \pmod{2}$, hence

$$A_n = \begin{cases} \{w : w^{r^n} = z_0\}, & n \text{ even}, \\ \{w : w^{r^n} = -z_0\}, & n \text{ odd}. \end{cases} \quad (69)$$

8.2 Branch histories and exact base- r coding

Let

$$\mathcal{H}_n := \{0, 1, \dots, r-1\}^n. \quad (70)$$

For a history $\mathbf{k} = (k_1, \dots, k_n)$, an unwrapped angle may be written as

$$\phi_j = \frac{\phi_{j-1} + \pi + 2\pi k_j}{r}, \quad \phi_0 = \text{Arg } z_0. \quad (71)$$

Iterating gives

$$\phi_n = \frac{\phi_0 + \pi S_n + 2\pi K_n}{r^n}, \quad (72)$$

where

$$K_n = k_1 r^{n-1} + k_2 r^{n-2} + \dots + k_{n-1} r + k_n. \quad (73)$$

As the digits vary, K_n runs through $0, 1, \dots, r^n - 1$.

Proposition 8.7 (History-endpoint bijection). *For $z_0 \neq 0$, the map from branch histories in $\mathcal{H}_n$ to the depth- n endpoints in A_n is bijective. In particular,*

$$|\mathcal{H}_n| = |A_n| = r^n. \quad (74)$$

Proof. The digit map $(k_1, \dots, k_n) \mapsto K_n$ is the standard base- r bijection onto $\{0, \dots, r^n - 1\}$. Distinct K_n give distinct angles modulo 2π because their differences are strictly smaller than r^n . $\square$

This exact bijection is special to the monomial model. In a more general multivalued dynamical system, different histories may collide at the same endpoint. That distinction becomes important when one moves away from $-z^r$.

9 Geometry of the complete branch sets

Write

$$z_0 = \rho_0 e^{i\theta_0}, \quad \rho_0 > 0. \quad (75)$$

Every point of A_n has the common modulus

$$R_n = \rho_0^{1/r^n}, \quad (76)$$

and adjacent arguments are separated by

$$\Delta_n = \frac{2\pi}{r^n}. \quad (77)$$

Theorem 9.1 (Regular-polygon geometry). *For $z_0 \neq 0$, the depth- n branch set is the vertex set of a regular r^n -gon centered at the origin, with circumradius R_n and angular mesh $2\pi/r^n$. Moreover*

$$R_n \rightarrow 1, \quad \Delta_n \rightarrow 0.$$

Proof. The closed depth equation identifies A_n with the complete set of r^n roots of a single nonzero complex number. Those roots have a common modulus and arguments differing by $2\pi/r^n$. $\square$

9.1 Exact Hausdorff distance

Let $N = r^n$. The one-sided distance from the polygon to the unit circle is exactly $|R_n - 1|$. The one-sided distance from the unit circle to the polygon is attained at the midpoint of an angular

gap.

Theorem 9.2 (Exact Hausdorff formula). *For $z_0 \neq 0$,*

$$d_H(A_n, \mathbb{S}^1) = \max \left\{ |R_n - 1|, \sqrt{R_n^2 + 1 - 2R_n \cos(\pi/r^n)} \right\}. \quad (78)$$

In particular $d_H(A_n, \mathbb{S}^1) = O(r^{-n})$.

Proof. For the first term, every vertex lies on the circle of radius R_n . For the second, a point of $\mathbb{S}^1$ is closest to one of the two adjacent vertices in its angular sector, and the worst case is the angular midpoint. The law of cosines then gives the displayed expression. $\square$

Theorem 9.3 (Sharp first-order Hausdorff asymptotic). *Let $L = \log \rho_0$. Then*

$$d_H(A_n, \mathbb{S}^1) = \frac{\sqrt{L^2 + \pi^2}}{r^n} + o(r^{-n}). \quad (79)$$

Equivalently,

$$r^n d_H(A_n, \mathbb{S}^1) \longrightarrow \sqrt{(\log \rho_0)^2 + \pi^2}. \quad (80)$$

Proof. Since $R_n = e^{L/r^n}$,

$$R_n - 1 = \frac{L}{r^n} + O(r^{-2n}),$$

so

$$|R_n - 1| = \frac{|L|}{r^n} + O(r^{-2n}).$$

For the angular term,

$$\begin{aligned} R_n^2 + 1 - 2R_n \cos(\pi/r^n) &= (R_n - 1)^2 + 2R_n(1 - \cos(\pi/r^n)) \\ &= \frac{L^2}{r^{2n}} + \frac{\pi^2}{r^{2n}} + o(r^{-2n}). \end{aligned}$$

Hence the angular term equals $r^{-n} \sqrt{L^2 + \pi^2} + o(r^{-n})$. Since $\sqrt{L^2 + \pi^2} > |L|$, the angular term eventually dominates the maximum. $\square$

The radial contribution therefore matters at the same first-order scale as the angular mesh when $\rho_0 \neq 1$. The frequently quoted approximation πr^{-n} is recovered exactly in the special case $\rho_0 = 1$.

9.2 Individual orbit asymptotics

The polygonal Hausdorff estimate has a deterministic analogue for a single principal orbit.

Theorem 9.4 (Sharp asymptotic for a principal orbit). *Let $c_n \in \mathcal{C}_r$ be the point of the attracting two-cycle having the same sign as the principal angle θ_n after any transient zero-angle event. Then*

$$r^n |z_n - c_n| \longrightarrow \sqrt{(\log \rho_0)^2 + \left(|\theta_0| - \frac{\pi}{r+1} \right)^2}. \quad (81)$$

Proof. The exact radial identity gives

$$\rho_n = \exp((\log \rho_0)/r^n) = 1 + \frac{\log \rho_0}{r^n} + O(r^{-2n}).$$

The exact absolute-angle formula gives

$$|\theta_n - \arg(c_n)| = \frac{1}{r^n} \left| |\theta_0| - \frac{\pi}{r+1} \right|.$$

Using

$$|\rho_n e^{i\theta_n} - e^{i\arg(c_n)}|^2 = (\rho_n - 1)^2 + 2\rho_n(1 - \cos(\theta_n - \arg(c_n)))$$

and $2(1 - \cos u) = u^2 + O(u^4)$ gives the result after multiplication by r^{2n} . $\square$

Corollary 9.5 (Uniform first-order convergence bound). *After the transient zero-angle event,*

$$\text{dist}(z_n, \mathcal{C}_r) \leq \frac{|\log \rho_0|}{r^n} \exp\left(\frac{|\log \rho_0|}{r^n}\right) + \frac{1}{r^n} \left| |\theta_0| - \frac{\pi}{r+1} \right|. \quad (82)$$

9.3 Convex hulls, areas, and edge scales

Let $N = r^n$. The convex hull of A_n is the regular N -gon with circumradius R_n , so

$$\text{Area}(\text{conv } A_n) = \frac{N}{2} R_n^2 \sin\left(\frac{2\pi}{N}\right). \quad (83)$$

Consequently

$$\text{Area}(\text{conv } A_n) \rightarrow \pi, \quad \ell_n = 2R_n \sin\left(\frac{\pi}{N}\right) \sim \frac{2\pi}{r^n}, \quad (84)$$

where ℓ_n is the edge length. Thus the same construction has two geometrically different limits: the vertex sets approximate the unit circle, while the convex hulls approximate the closed unit disk.

10 Normalized branch measures and weighted branch measures

The finite branch sets carry more information than their geometry alone. Once a probability is assigned to each branch, the inverse tree becomes a random object, and the regular polygons acquire limiting probability measures.

10.1 Uniform branch selection

Define

$$\mu_n := \frac{1}{r^n} \sum_{z \in A_n} \delta_z, \quad (85)$$

and let $m_{\mathbb{S}^1}$ be normalized Haar probability measure on $\mathbb{S}^1$.

Theorem 10.1 (Weak convergence to Haar measure). *If $z_0 \neq 0$, then*

$$\mu_n \rightharpoonup m_{\mathbb{S}^1} \quad (86)$$

weakly as probability measures on $\mathbb{C}$.

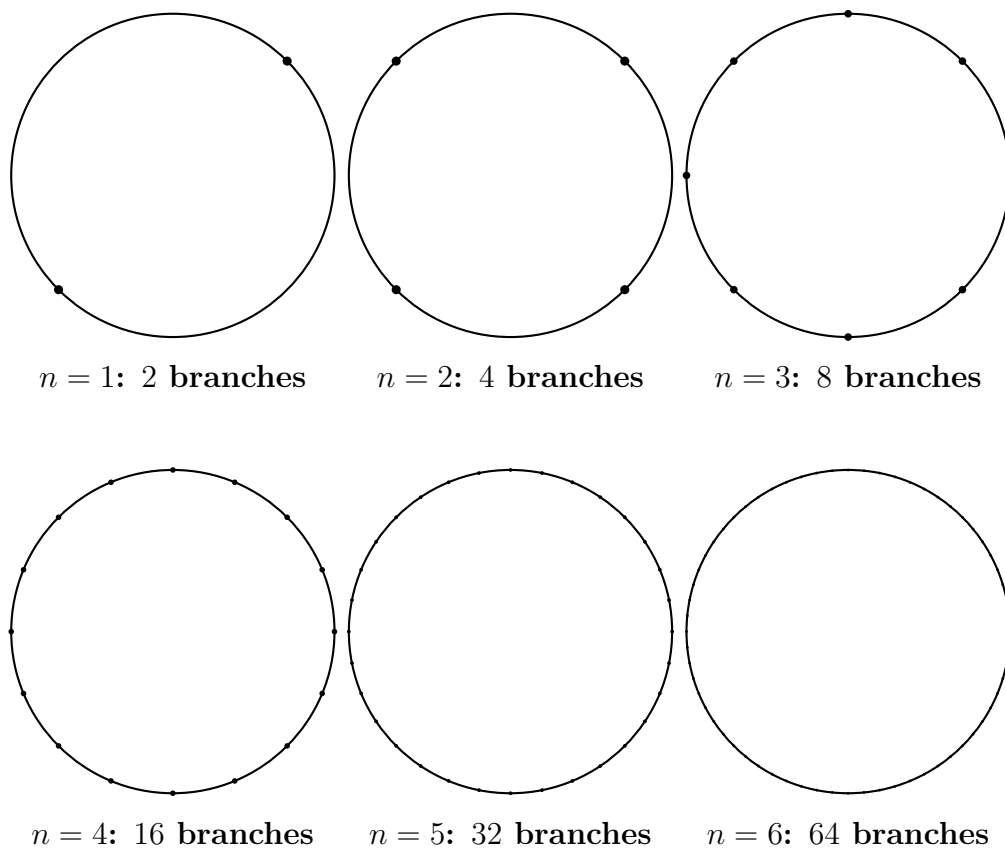


Figure 1: The complete branch sets in the square-root case for a unit-modulus initial point. The number of endpoints doubles at every depth, and the endpoints become an increasingly fine polygon on the unit circle.

Proof. Write $N = r^n$ and

$$A_n = \{R_n e^{i(\phi_n + 2\pi k/N)} : 0 \leq k < N\}.$$

For any fixed nonzero integer ℓ , the ℓ th Fourier coefficient of the angular counting measure is

$$\frac{1}{N} \sum_{k=0}^{N-1} e^{i\ell(\phi_n + 2\pi k/N)} = e^{i\ell\phi_n} \frac{1}{N} \sum_{k=0}^{N-1} e^{2\pi i\ell k/N}.$$

Once $N > |\ell|$, the geometric sum vanishes identically. Thus every fixed nonzero Fourier coefficient is eventually exactly zero. Trigonometric polynomials are uniformly dense in $C(\mathbb{S}^1)$, and $R_n \rightarrow 1$, so the normalized polygon measures converge weakly to Haar measure. $\square$

Remark 10.2. *The result is stronger than ordinary equidistribution in the usual asymptotic sense: each fixed nonzero Fourier mode becomes exactly zero after a finite depth.*

10.2 Weighted branch probabilities and chronological reversal

Let

$$p = (p_0, \dots, p_{r-1}), \quad p_k \geq 0, \quad \sum_{k=0}^{r-1} p_k = 1. \quad (87)$$

For a chronological history $\mathbf{k} = (k_1, \dots, k_n)$ define

$$p(\mathbf{k}) = \prod_{j=1}^n p_{k_j}.$$

The weighted endpoint measure is

$$\mu_n^{(p)} := \sum_{\mathbf{k} \in \mathcal{H}_n} p(\mathbf{k}) \delta_{\Phi_n(\mathbf{k})}. \quad (88)$$

Let $(K_j)_{j \geq 1}$ be i.i.d. digits with $\mathbb{P}(K_j = k) = p_k$. From the chronological branch formula,

$$\phi_n = \frac{\text{Arg } z_0}{r^n} + \frac{\pi S_n}{r^n} + 2\pi \sum_{j=1}^n \frac{K_j}{r^{n-j+1}}. \quad (89)$$

The final sum is the finite base- r expansion read in reverse chronological order. It is therefore important not to identify it pathwise with the usual forward partial sum.

Define instead

$$X_n := \sum_{j=1}^n \frac{K_j}{r^j}, \quad X := \sum_{j=1}^{\infty} \frac{K_j}{r^j}. \quad (90)$$

Because $(K_1, \dots, K_n)$ has the same joint distribution as $(K_n, \dots, K_1)$, the reversed finite sum in (89) has exactly the same distribution as X_n . This is an equality in distribution, not a pathwise equality.

Theorem 10.3 (Weighted branch-measure limit). *For every probability vector p and every $z_0 \neq 0$,*

$$\mu_n^{(p)} \rightharpoonup \nu_p, \quad (91)$$

where ν_p is the law of

$$Y_p := \exp\left(i\left(2\pi X + \frac{\pi}{r-1}\right)\right). \quad (92)$$

In particular, the limit is independent of z_0 except through a vanishing r^{-n} transient.

Proof. Since $S_n/r^n \rightarrow 1/(r-1)$, $\text{Arg } z_0/r^n \rightarrow 0$, and $X_n \rightarrow X$ almost surely, the reversed finite angular sum converges in distribution to the angle in (92). The radial factor is $R_n = \rho_0^{1/r^n} \rightarrow 1$ deterministically. For every bounded continuous test function f on $\mathbb{C}$, the random variables $f(R_n e^{i\phi_n})$ are bounded and converge in distribution to $f(Y_p)$; equivalently, by using the reversed digit vector and the almost sure convergence of X_n , one obtains convergence of expectations. This proves weak convergence. $\square$

Corollary 10.4 (Information growth). *Let*

$$h(p) := -\sum_{k=0}^{r-1} p_k \log p_k. \quad (93)$$

Then

$$H(K_1, \dots, K_n) = nh(p), \quad \log |\mathcal{H}_n| = n \log r. \quad (94)$$

Thus uniform branch selection maximizes information growth, while biasing the branch probabilities lowers the information produced per recursive level.

10.3 The self-similar structure of the limit measure

Let $S_k : [0, 1] \rightarrow [0, 1]$ be

$$S_k(x) = \frac{x+k}{r}, \quad k = 0, \dots, r-1. \quad (95)$$

Then the law η_p of X is the unique self-similar probability measure satisfying

$$\eta_p = \sum_{k=0}^{r-1} p_k (S_k)_\# \eta_p. \quad (96)$$

The map

$$\Psi(x) = e^{i(2\pi x + \pi/(r-1))} \quad (97)$$

then gives $\nu_p = \Psi_\# \eta_p$.

The only possible nonuniqueness in base- r coding occurs at the familiar countable set of endpoints with two expansions. This finite-to-one identification does not alter the Hausdorff dimension, and it does not affect the almost-everywhere local-dimension calculations below.

Theorem 10.5 (Exact dimensionality and dimension formula). *For every probability vector p ,*

$$\dim_H \nu_p = \frac{h(p)}{\log r}. \quad (98)$$

More precisely, ν_p is exact dimensional: for ν_p -almost every x ,

$$\lim_{\varepsilon \downarrow 0} \frac{\log \nu_p(B(x, \varepsilon))}{\log \varepsilon} = \frac{h(p)}{\log r}. \quad (99)$$

def

Proof. For a symbolic sequence $\omega = (\omega_1, \omega_2, \dots)$, the cylinder of length n has diameter comparable to r^{-n} , while its mass is

$$\eta_p([\omega_1 \cdots \omega_n]) = \prod_{j=1}^n p_{\omega_j}.$$

For η_p -almost every sequence, the strong law of large numbers gives

$$\frac{1}{n} \log \eta_p([\omega_1 \cdots \omega_n]) \longrightarrow \sum_k p_k \log p_k = -h(p).$$

Since the cylinder diameter has logarithm $-n \log r + O(1)$, the local dimension is $h(p)/\log r$. Endpoint identifications form a countable exceptional set and the circle map Ψ is Lipschitz, so the same almost-everywhere dimension holds for ν_p . Thus the measure is exact dimensional and its Hausdorff dimension is the displayed value. $\square$

Remark 10.6. *The dimension formula is a classical Bernoulli/self-similar phenomenon. In the present paper the important point is that the root-tree construction realizes it exactly and transparently; no additional thermodynamic machinery is required.*

Corollary 10.7 (Haar characterization). *The following are equivalent:*

- (i) $p_k = 1/r$ for all k ;
- (ii) $h(p) = \log r$;
- (iii) $\nu_p = m_{\mathbb{S}^1}$.

If p is nonuniform, then $h(p) < \log r$, so ν_p has Hausdorff dimension < 1 and is singular with respect to Haar measure.

10.4 Multifractal frequency classes

The uniform exact-dimensional statement can be refined by conditioning on symbolic frequencies. Let $q = (q_0, \dots, q_{r-1})$ be a probability vector, and let $E(q)$ denote the set of sequences whose digit frequencies converge to q .

For $x \in E(q)$ and $p_k > 0$ on the support of q , the cylinder masses satisfy

$$\frac{1}{n} \log \mu_p([\omega_1 \cdots \omega_n]) \longrightarrow \sum_k q_k \log p_k.$$

Hence the local dimension of ν_p along the frequency class is

$$\boxed{d_{\nu_p}(x) = \frac{-\sum_k q_k \log p_k}{\log r}.} \tag{100}$$

At the same time, the Hausdorff dimension of the frequency class itself is

$$\boxed{\dim_H E(q) = \frac{-\sum_k q_k \log q_k}{\log r}.} \tag{101}$$

This gives an explicit multifractal description of the biased branch tree. The typical frequency $q = p$ recovers the exact-dimensional value $h(p)/\log r$.

10.5 Uniformity, singularity, and the geometry of bias

When all digits are equally likely, the coding map produces Lebesgue measure on the base- r interval and hence Haar measure on the circle. When the digit probabilities are nonuniform but all are positive, the support remains the entire circle, yet the measure is singular and assigns exponentially different masses to cylinders of equal geometric size. Thus the inverse tree separates two notions of uniformity: geometric uniformity of endpoint placement and probabilistic uniformity of branch selection.

This distinction is one of the reasons to treat branch geometry and branch probability as separate mathematical layers.

11 A detailed square-root case study

The order-2 map is

$$F_2(z) = \sqrt{-z}. \quad (102)$$

Its principal two-cycle is

$$\{e^{i\pi/3}, e^{-i\pi/3}\}. \quad (103)$$

11.1 The orbit beginning at $z_0 = i$

Here $\rho_0 = 1$ and $\theta_0 = \pi/2$. Formula (25) gives

$$\alpha_n = \frac{\pi}{3} + \left(-\frac{1}{2}\right)^n \frac{\pi}{6}. \quad (104)$$

Since the initial angle is not the exceptional value π , the signs alternate from the first step onward. Hence

$$\boxed{\theta_n = (-1)^n \frac{\pi}{3} + \frac{\pi}{6 \cdot 2^n}}. \quad (105)$$

The first several values are

$$\begin{array}{lll} \theta_0 = \frac{\pi}{2}, & \theta_1 = -\frac{\pi}{4}, & \theta_2 = \frac{3\pi}{8}, \\ \theta_3 = -\frac{5\pi}{16}, & \theta_4 = \frac{11\pi}{32}, & \theta_5 = -\frac{21\pi}{64}. \end{array}$$

Because $\rho_n = 1$, the entire orbit lies on the unit circle and the exact complex value is

$$z_n = \exp\left(i \left[(-1)^n \frac{\pi}{3} + \frac{\pi}{6 \cdot 2^n}\right]\right). \quad (106)$$

11.2 Changing the initial modulus

For $z_0 = \rho_0 e^{i\theta_0}$,

$$z_n = \rho_0^{1/2^n} e^{i\theta_n}. \quad (107)$$

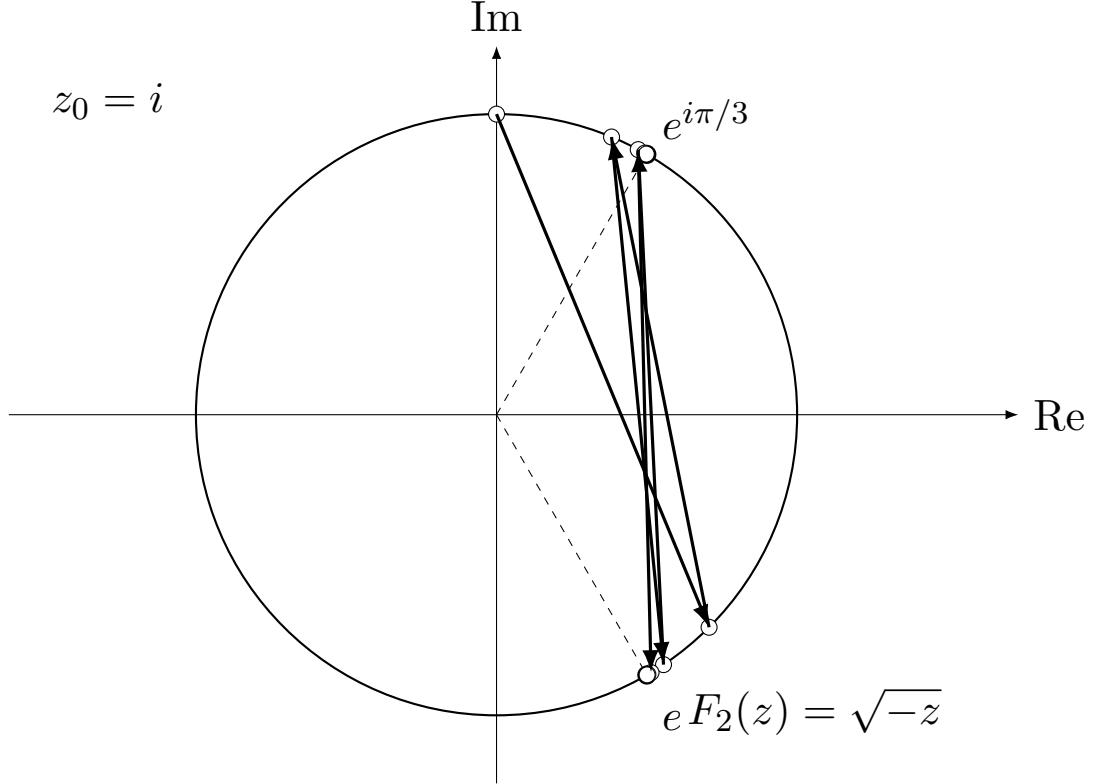


Figure 2: The principal orbit of $\sqrt{-z}$ starting at i . The radius is identically one and the angular points alternate around the attracting pair $e^{\pm i\pi/3}$.

For instance,

$$\begin{aligned} |F_2^n(4i)| &= 4^{1/2^n} \downarrow 1, \\ |F_2^n(i/4)| &= (1/4)^{1/2^n} \uparrow 1. \end{aligned}$$

The phase is independent of the initial modulus.

12 Variable root orders and nonautonomous contraction

The fixed-order system has an exact nonautonomous analogue in which the root order may change from one stage to the next:

$$z_{n+1} = \text{Root}_{r_{n+1}}^{\text{pr}}(-z_n), \quad r_{n+1} \geq 2. \quad (108)$$

Define the cumulative product

$$R_n := \prod_{j=1}^n r_j, \quad R_0 := 1. \quad (109)$$

Because each $r_j \geq 2$,

$$R_n \geq 2^n \rightarrow \infty. \quad (110)$$

Thus the basic forgetting mechanism is unconditional under the stated root-order assumption.

12.1 Exact radial law

Proposition 12.1 (Variable-order radial law). *For every nonzero branch value generated at depth n ,*

$$|z_n| = |z_0|^{1/R_n}. \quad (111)$$

Consequently $|z_n| \rightarrow 1$.

Proof. At step j , taking an r_j th root divides the logarithm of the modulus by r_j . Summing these divisions gives the product R_n . $\square$

12.2 Exact principal absolute-angle recurrence

Let $\alpha_n = |\text{Arg } z_n|$. The sign-reversal mechanism is unchanged by the varying order, giving

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r_{n+1}}. \quad (112)$$

Theorem 12.2 (Exact variable-order angle formula). *For every $n \geq 1$,*

$$\alpha_n = \frac{(-1)^n \alpha_0}{R_n} + \pi \sum_{j=1}^n \frac{(-1)^{n-j}}{r_j r_{j+1} \cdots r_n}. \quad (113)$$

Moreover, for any two initial magnitudes $\alpha_0, \tilde{\alpha}_0$ driven by the same order sequence,

$$|\alpha_n - \tilde{\alpha}_n| = \frac{|\alpha_0 - \tilde{\alpha}_0|}{R_n} \leq 2^{-n} |\alpha_0 - \tilde{\alpha}_0|. \quad (114)$$

Proof. The first formula follows by induction from (112); the first two cases are

$$\alpha_1 = -\frac{\alpha_0}{r_1} + \frac{\pi}{r_1}, \quad \alpha_2 = \frac{\alpha_0}{r_1 r_2} - \frac{\pi}{r_1 r_2} + \frac{\pi}{r_2}.$$

The general expression follows by continuing the same substitution. Subtracting two solutions removes the forcing terms and leaves multiplication by $-1/r_{n+1}$ at each step, giving the exact forgetting identity. $\square$

The estimate shows more than convergence: every pair of initial angular magnitudes is exponentially synchronized, uniformly over all admissible order sequences.

12.3 Periodic order sequences

Suppose $r_{n+p} = r_n$. The p -step absolute-angle map is affine with slope

$$\lambda_p = \frac{(-1)^p}{r_1 r_2 \cdots r_p}, \quad |\lambda_p| < 1. \quad (115)$$

Theorem 12.3 (Attracting periodic magnitude profile). *If (r_n) is p -periodic, then there exists a unique p -periodic magnitude profile*

$$\alpha_0^*, \dots, \alpha_{p-1}^*, \quad \alpha_{n+p}^* = \alpha_n^*,$$

and every solution converges to it. The p -step contraction factor is

$$(r_1 r_2 \cdots r_p)^{-1}.$$

If no profile value vanishes, sign alternation determines the signed period from the least period q of the magnitude profile: it is q when q is even and $2q$ when q is odd, apart from the exceptional zero-angle case.

Proof. The composition of one full order period is an affine contraction of the compact interval $[0, \pi]$ with absolute slope strictly smaller than one. It therefore has a unique fixed point. Reconstructing the intermediate values produces the unique periodic profile. The signed orbit changes sign at each nonzero step, so a return of the magnitude after an even number of steps preserves the sign while an odd return reverses it; this gives the stated period rule. $\square$

12.4 Mixed-order multivalued algebra

For the fully branched recursion, let every point at depth $j - 1$ generate every r_j th root of its negation. Put

$$E_n := 1 + r_1 + r_1 r_2 + \cdots + r_1 r_2 \cdots r_{n-1}. \quad (116)$$

Theorem 12.4 (Exact mixed-order branch equation). *For $z_0 \neq 0$,*

$$\boxed{A_n = \left\{ w : w^{R_n} = (-1)^{E_n} z_0 \right\}}. \quad (117)$$

Hence

$$|A_n| = R_n.$$

Proof. At depth one, $w^{r_1} = -z_0$. Suppose the depth- n equation has the stated form. If $w^{r_{n+1}} = -z$, then

$$w^{R_{n+1}} = (-z)^{R_n} = (-1)^{R_n} z^{R_n} = (-1)^{R_n + E_n} z_0.$$

Since $E_{n+1} = E_n + R_n$, the induction closes. $\square$

The mixed-order result shows that the regular-polygon phenomenon is not tied to a constant root order. What changes is the polygon cardinality, now equal to the cumulative product R_n .

12.5 A stronger nonautonomous interpretation

The variable-order system separates into two roles. The product R_n governs the rate at which all dependence on the initial radius and initial angular magnitude disappears, while the alternating forcing term in the exact formula determines the residual magnitude profile. Thus the order sequence acts simultaneously as a contraction schedule and as a forcing schedule.

This distinction is useful when comparing deterministic principal dynamics with the full branch tree. In the principal orbit, the signs are determined by the branch convention and the absolute magnitudes synchronize. In the full tree, the same cumulative product controls the number of endpoints and their angular mesh. The two descriptions therefore remain different, but are governed by the same multiplicative depth scale.

13 Infinite radicals: four non-equivalent meanings

A formal infinite radical does not determine a unique mathematical object until the finite approximants and limiting procedure are specified. The present recursion makes this ambiguity particularly transparent because several natural constructions coexist and have different limits.

Theorem 13.1 (Four asymptotic objects). *For $z_0 \neq 0$ and $F_r(z) = \text{Root}_r^{\text{pf}}(-z)$:*

- (i) *the fixed-point equation $L = F_r(L)$ has exactly one solution, $L = 0$;*
- (ii) *the deterministic principal orbit has no pointwise limit, but its ω -limit set is exactly $\mathcal{C}_r = \{e^{\pm i\pi/(r+1)}\}$;*
- (iii) *the full depth- n branch sets converge to $\mathbb{S}^1$ in Hausdorff distance, with the sharp asymptotic (79);*
- (iv) *the normalized counting measures converge weakly to Haar measure, while a nonuniform branch distribution gives the exact-dimensional measure described in Section 98.*

Table 1: Four mathematically distinct meanings of infinite depth.

Construction	Limiting object
Fixed-point equation $L = F_r(L)$	$L = 0$ only
Principal iteration from $z_0 \neq 0$	no pointwise limit; $\omega(z_0) = \mathcal{C}_r$
All branches through depth n	$A_n \rightarrow \mathbb{S}^1$ in Hausdorff distance
Uniform branch counting	$\mu_n \rightharpoonup m_{\mathbb{S}^1}$
Weighted branch probabilities	$\mu_n^{(p)} \rightharpoonup \nu_p$

Interpretive principle. The statement “the infinite radical equals L ” is incomplete unless the intended construction is specified. A fixed-point limit, a deterministic orbit, a set-valued inverse limit, and a measure-valued limit are distinct mathematical objects.

13.1 Fixed-point meaning

The fixed-point equation is algebraically legitimate:

$$L = \text{Root}_r^{\text{pf}}(-L).$$

The only solution is $L = 0$. This does not describe the asymptotic behavior of a nonzero principal orbit because 0 is not approached by such an orbit in the radial coordinate; nonzero radii instead converge to 1.

13.2 Pointwise iterative meaning

For every nonzero z_0 , the principal orbit approaches the two-cycle $\mathcal{C}_r$. Thus the infinite nested expression understood as a pointwise limit of its principal finite approximants is generally not convergent. The correct deterministic statement is asymptotic periodicity rather than convergence to a single complex number.

13.3 Set-valued meaning

Retaining all branches produces the increasing family of finite sets A_n . These do not converge pointwise, because their cardinality grows, but they converge in the Hausdorff metric to the unit circle. The corresponding convex hulls converge instead to the closed unit disk.

13.4 Measure-valued meaning

Assigning branch probabilities produces a sequence of probability measures $\mu_n^{(p)}$. Uniform branch selection yields Haar measure; biased selection yields an exact-dimensional self-similar measure of dimension $h(p)/\log r$. The measure-valued interpretation therefore contains information invisible to either the deterministic orbit or the mere geometric closure of the branch sets.

13.5 Order of limits

A related source of confusion is the distinction between changing the root order and increasing recursion depth. For example,

$$\text{Root}_n^{\text{pr}}(i) = e^{i\pi/(2n)} \rightarrow 1,$$

while repeated application of a fixed negative square root produces a nontrivial attracting two-cycle. Likewise, taking all n th roots of a fixed complex number gives n points, whereas taking all branches through n recursive stages of a fixed r th root gives r^n points. These are different limiting procedures and should never be merged into a single notation without an explicit convention.

14 Branch-cut hitting and exceptional initial data

Because the principal map is discontinuous on the positive real axis, it is useful to ask which orbits can actually land on that discontinuity.

Proposition 14.1 (The branch-cut hitting set). *For the fixed-order negative-root map F_r , a nonzero principal orbit can hit the positive real axis only if its initial point lies on the negative real axis. In that exceptional case the orbit passes through the positive real axis exactly once and then leaves it.*

Proof. In angular coordinates, landing on the positive real axis means $\theta_n = 0$. From (12), $T_r(\theta) = 0$ occurs only at $\theta = \pi$. After one iteration every angle satisfies $|\theta| \leq \pi/r < \pi$, so no later iterate can equal π . Therefore the only possible hit is at the first step from $\theta_0 = \pi$, i.e. the negative real axis. $\square$

This observation is small but conceptually useful: the global discontinuity is dynamically relevant, yet in the negative-root system the attracting cycle itself remains uniformly separated from the discontinuity.

15 Relations with classical complex dynamics

The polynomial

$$P_r(z) = -z^r \tag{118}$$

has a particularly simple forward action on the unit circle:

$$P_r(e^{i\theta}) = e^{i(r\theta+\pi)}. \quad (119)$$

Thus, modulo a rigid rotation, the restriction to $\mathbb{S}^1$ is the standard degree- r expanding map on the circle. The finite branch sets studied earlier are its exact inverse-image levels. In this language, the regular polygons arise because the inverse image of one point under a monomial is again a rigidly translated set of equally spaced roots.

The broad theory of iteration of rational and polynomial maps is classical; standard references include Beardon and Milnor.⁴

The multivalued viewpoint also belongs naturally to the theory of holomorphic correspondences. Recent work of Hemmingsson develops equidistribution results for certain holomorphic correspondences and associated invariant measures.⁵

Nonautonomous monomial iteration is also an active topic. A recent 2026 preprint of Pathirana, Petersen, and Vardakis studies Julia sets for sequences of monomials; this is relevant to the variable-order direction of the present paper, although the principal-branch contraction studied here is a more specialized inverse-root construction.⁶

16 What is special about the monomial model?

The exact solvability of this paper is not accidental. It is a consequence of the monomial structure. It is therefore useful to state explicitly which features are special.

Property	$-z^r$	az^r	General polynomial
Exact common modulus at inverse depth	yes	yes	generally no
Regular polygon inverse sets	yes	yes	generally no
Exact base- r branch coding	yes	yes	generally no
Fourier cancellation at finite depth	yes	yes	generally no
Exact principal angular affine law	yes	phase-shifted	generally no
Simple weighted digit model	yes	yes	generally no

This comparison is not merely a disclaimer. It identifies a natural research program: determine which portions of the theory survive under perturbation and which are rigid signatures of the monomial case.

⁴Beardon's book gives a systematic account of iteration of rational functions, including periodic points, invariant sets, critical points, and the role of expanding maps. Milnor's lectures provide a widely used introduction to one-complex-variable dynamics. The present paper uses only a small and elementary portion of that broader theory. See Beardon [12] and Milnor [13].

⁵The comparison is conceptual rather than a claim that the present monomial correspondence requires Hemmingsson's machinery: here the limiting measures can be computed directly from finite geometric sums. See Hemmingsson [10].

⁶Because the nonautonomous literature is evolving, the comparison should be kept precise: the mixed-order formulas here are exact algebraic identities for inverse root sets, while general nonautonomous monomial Julia-set questions concern a substantially broader class of forward systems. See Pathirana, Petersen, and Vardakis [11].

17 Further structural directions and open problems

The results above suggest several problems that are substantially more difficult than the exactly solvable core.

17.1 Complete phase-parameter classification

The periodic-itinerary theorem reduces periodicity of $F_{r,\beta}$ to a finite admissibility problem. A natural target is a complete classification of the (β, p) for which a strictly admissible period- p word exists. Because each branch is contracting, the real difficulty is combinatorial rather than differential: one must understand which finite integer words survive the principal-value inequalities.

Conjecture 17.1 (Finite-itinerary completeness). *For fixed $r \geq 2$ and $\beta \in (-\pi, \pi]$, every nonwandering periodic orbit of the principal phase map $T_{r,\beta}$ is represented by a strictly admissible finite branch word or by an explicitly classifiable boundary itinerary. In particular, the periodic dynamics should be reducible to a finite symbolic admissibility problem on every compact parameter interval avoiding branch-boundary collisions.*

The conjecture is deliberately stated as a program rather than claimed as a theorem of this paper.

17.2 Higher-period phase windows

The period-three example at $r = 2$, $\beta = 2\pi/3$ suggests a rich but highly structured parameter picture. It would be natural to determine explicit windows of phases supporting periods 3, 4, 5, $\dots$ and to characterize their endpoints by branch-boundary equalities.

17.3 Weighted multifractal limits

For a nonuniform probability vector p , the measure ν_p is a natural candidate for a dimension-theoretic analysis. In the standard Bernoulli setting one expects its exact dimension to be controlled by

$$\frac{h(p)}{\log r}, \quad (120)$$

with $h(p)$ the Shannon entropy, subject to the usual treatment of base- r endpoint ambiguities. A complete development would place the branch-tree measures of the present paper inside the theory of self-similar and exact-dimensional measures.

17.4 Perturbed maps

A natural nonlinear extension is

$$G(z) = \text{Root}_r^{\text{Pf}}(-az + b), \quad a, b \in \mathbb{C}. \quad (121)$$

The fixed-point equation becomes

$$z^r = -az + b, \quad (122)$$

and the branch boundary is the inverse image under $-az + b$ of the negative real axis. The resulting angular map is no longer globally affine, so the exact contraction theory of the present

paper becomes a model rather than a final answer.

17.5 Branch collisions

For the monomial model, branch histories and endpoints are in bijection at each fixed depth. In perturbed systems this need not remain true. One may define the collision ratio

$$C_n := \frac{\#\{\text{branch histories of length } n\}}{\#\{\text{distinct endpoints at depth } n\}} \quad (123)$$

and ask how C_n behaves under perturbation. This provides a concrete measure of how much recursive history is retained by the endpoint geometry.

17.6 Nonautonomous principal dynamics

For arbitrary (r_n) , the exact formula (113) gives strong control of the dependence on the initial condition, but the asymptotic signed itinerary can depend delicately on the order sequence. Periodic, eventually periodic, random, and slowly varying root orders should be treated separately.

17.7 Order of limits

There are two independent parameters: recursive depth n and root order r . For the principal negative-root cycle,

$$\frac{\pi}{r+1} = O(r^{-1}), \quad (124)$$

while the depth- n radial and angular errors are $O(r^{-n})$. Understanding asymptotic regimes in which $n = n(r)$ grows with r produces a useful two-scale limit problem.

17.8 Other metrics

The exact Hausdorff formula suggests studying Wasserstein distance, bounded-Lipschitz distance, angular discrepancy, and optimal matching between A_n and $\mathbb{S}^1$. Because the finite sets are regular polygons, these questions remain unusually tractable and may admit sharp constants rather than merely big- O estimates.

18 Contracted-rotation normal form and arithmetic phase classes

The normalization in Section 7 deserves a fuller treatment because it identifies the phase family with a classical one-dimensional dynamical family without sacrificing the exact complex-root origin of the map. The resulting normal form also gives a useful route to convergence statements for substantial classes of parameters.

18.1 The circle map and a canonical lift

Let

$$\lambda := \frac{1}{r}, \quad \delta := \delta_\beta = \frac{r-1}{2r} + \frac{\beta}{2\pi}. \quad (125)$$

Then the normalized angular map is

$$f_{\lambda,\delta}(x) = \{\lambda x + \delta\}, \quad x \in [0, 1). \quad (126)$$

For the parameter range $\beta \in (-\pi, \pi]$ and $r \geq 2$, one has

$$0 \leq \delta < 1, \quad 0 < \lambda < 1. \quad (127)$$

More precisely, the lower endpoint is approached but not included when $r = 2$, while for $r > 2$ the offset remains uniformly positive.

A canonical lift is

$$F_{\lambda, \delta}(x) := \lambda\{x\} + \delta + \lfloor x \rfloor, \quad x \in \mathbb{R}, \quad (128)$$

which satisfies

$$F_{\lambda, \delta}(x+1) = F_{\lambda, \delta}(x) + 1, \quad \{F_{\lambda, \delta}(x)\} = f_{\lambda, \delta}(\{x\}). \quad (129)$$

The associated rotation number is classically defined by

$$\rho(\lambda, \delta) := \lim_{n \rightarrow \infty} \frac{F_{\lambda, \delta}^{\circ n}(x)}{n}, \quad (130)$$

where the limit exists and is independent of the initial lift coordinate in the standard contracted-rotation setting [17].

The number ρ records the asymptotic average integer winding of the lifted orbit. It is therefore complementary to the finite branch index k_j used in the principal-angle recurrence: the latter records the local principal-value reduction, while ρ records the long-term average modular displacement.

18.2 Recovering the fixed-point regime from the normal form

The fixed-point condition in the normalized coordinate can be examined directly. If there is no modular wrap, then

$$x = \lambda x + \delta,$$

so

$$x_* = \frac{\delta}{1 - \lambda}. \quad (131)$$

The condition $x_* \in [0, 1)$ is equivalent to

$$0 \leq \delta < 1 - \lambda. \quad (132)$$

Substituting $\lambda = 1/r$ and the definition of δ gives precisely the phase threshold obtained in the direct angular calculation.

In the original angle variable, this is the fixed point

$$\theta_* = \frac{\beta}{r - 1}.$$

Thus the two descriptions agree exactly:

Complex angle	normalized coordinate	parameter condition
$\theta_* = \beta/(r - 1)$	$x_* = \delta/(1 - 1/r)$	$ \beta \leq \pi - \pi/r$

The significance of this calculation is that the fixed-point threshold is not an artifact of a particular coordinate system. It is the geometric point at which the affine fixed point reaches the discontinuity of the modular reduction.

18.3 Periodic orbits and rotation number

Suppose the normalized orbit is periodic with least period p . Let the lift satisfy

$$F_{\lambda,\delta}^{\circ p}(x) = x + m \quad (133)$$

for some integer m . Then

$$\rho = \frac{m}{p}. \quad (134)$$

Thus every periodic orbit has rational rotation number.

Conversely, standard contracted-rotation theory implies that for a fixed contracted rotation there is at most one periodic orbit when the rotation number is rational; when the rotation number is irrational, the limit set is a Cantor set on which the map is topologically conjugate to an irrational rotation [17]. The distinction is particularly useful here because it prevents a misleading interpretation of the many symbolic words in Section 7: distinct chronological words may represent cyclic relabelings of the same geometric periodic orbit.

Proposition 18.1 (Uniqueness at a fixed phase). *For a fixed pair (r, β) , the phase-family map has at most one periodic orbit in the standard contracted-rotation dynamics. Whenever such an orbit is present, it is attracting along its periodic branch itinerary with multiplier r^{-p} , where p is its least period.*

Proof. The normalized map is a contracted rotation with $0 < \lambda < 1$. The uniqueness of a periodic orbit for rational rotation number is a standard property of this family [17]. On any strictly admissible periodic itinerary, the derivative of the p -step affine branch is $\lambda^p = r^{-p}$. $\square$

This proposition should be read together with the exact window theorem. A single phase parameter may lie inside one periodic window even though many symbolic descriptions are possible through cyclic shifts or alternative word representatives. The geometric periodic orbit remains unique.

18.4 Arithmetic phase parameters

The relation

$$\delta_\beta = \frac{r-1}{2r} + \frac{\beta}{2\pi} \quad (135)$$

allows arithmetic information about contracted rotations to be transferred directly to the root family.

Theorem 18.2 (Rational phase parameters yield periodic normalized dynamics). *Suppose $\beta/\pi \in \mathbb{Q}$. Then $\delta_\beta \in \mathbb{Q}$ and $\lambda = 1/r \in \mathbb{Q}$. Hence the normalized phase map has rational rotation number. Consequently its limiting dynamics are periodic rather than an irrational Cantor rotation.*

Proof. The rationality of λ and δ_β is immediate. Laurent and Nogueira prove that algebraic parameters for the standard contracted-rotation family give a rational rotation number [17]. Since rational parameters are algebraic, the conclusion follows. For rational rotation number, the limit set is the periodic orbit in the standard contracted-rotation classification. $\square$

The theorem gives a substantial convergence class beyond the elementary fixed-point regime. It implies that every rational phase ratio β/π produces asymptotically periodic normalized

angular dynamics, even when the principal map repeatedly crosses its discontinuity before settling onto its periodic attractor.

Corollary 18.3 (Algebraic phase ratio). *If β/π is algebraic, then δ_β is algebraic, and the normalized contracted rotation has rational rotation number. Thus algebraic phase ratios also belong to the periodic arithmetic class described above.*

Proof. Again $\lambda = 1/r$ is algebraic and δ_β is algebraic. Apply the algebraic-parameter theorem for contracted rotations [17]. $\square$

The preceding corollary is deliberately stated at the level of the normalized angular map. Recovering the exact complex periodic orbit still requires the admissibility calculation in the original angle coordinate.

18.5 The irrational regime as a conditional alternative

The contracted-rotation framework also identifies what must happen if the rotation number is irrational. In that case the limiting set is a Cantor set and the dynamics on it are topologically conjugate to an irrational circle rotation [17]. Hence the present phase family has a principled nonperiodic alternative, but this alternative is tied to the arithmetic value of the rotation number, not to the mere presence of a branch discontinuity.

This distinction corrects a common informal narrative. A discontinuous contraction need not be chaotic in the positive-entropy sense; it may instead support a minimal Cantor system with zero entropy and irrational rotational order. Conversely, the existence of such behavior in the broader contracted-rotation class does not establish its occurrence for every root-induced parameter. The exact parameterization of δ_β remains the object to analyze.

18.6 A phase diagram in the $r = 2$ case

For $r = 2$,

$$\lambda = \frac{1}{2}, \quad \delta_\beta = \frac{1}{4} + \frac{\beta}{2\pi}. \quad (136)$$

The fixed-point regime is

$$|\beta| \leq \frac{\pi}{2}.$$

The parameter $\beta = 2\pi/3$ lies beyond this interval and produces the exact period-three orbit calculated earlier. The parameter $\beta = \pi$ is the negative-root endpoint and produces the rigid two-cycle.

The following table organizes several representative phases.

The table is not intended as a complete classification. Its role is to show that the transition from fixed points to higher periodicity occurs inside one explicitly normalized one-parameter family.

18.7 The symbolic and rotational descriptions are complementary

The symbolic description records exactly where the principal-value operation subtracts an integer multiple of 2π . The contracted-rotation description packages the same information into a circle map and its lift. Neither description subsumes the other completely.

Table 2: Representative phase regimes for $r = 2$.

Phase β	Rotation-type description	Principal angular behavior
0	fixed point	$\theta_n \rightarrow 0$
$\pi/2$	threshold fixed point	affine fixed point reaches the wedge boundary
$2\pi/3$	rational periodic regime	attracting period-three orbit
π	symmetric negative phase	attracting period-two orbit at $\pm\pi/3$

The symbolic description is preferable for exact algebraic calculations such as

$$\theta_0 = \frac{\beta S_p - 2\pi K_p^{\text{chron}}}{r^p - 1},$$

whereas the rotation-number viewpoint is preferable for discussing long-term winding, parameter arithmetic, and the conditional Cantor alternative. Keeping the two descriptions separate but linked is therefore part of the conceptual architecture of the paper.

18.8 A precise boundary statement

A periodic orbit inside a strict window remains uniformly separated from the discontinuity set. At an interior parameter endpoint, at least one branch expression satisfies

$$\theta_j + \beta - 2\pi k_j = \pm\pi.$$

The equality at $-\pi$ is excluded by the principal-value convention, whereas the equality at $+\pi$ is included. This asymmetry means that a boundary parameter cannot be inferred merely from the geometric size of the window; the side from which the orbit reaches the discontinuity also matters.

The contracted-rotation normal form makes this visible geometrically: the periodic orbit collides with the modular cut in the normalized coordinate exactly when the affine image reaches an endpoint of the chosen fundamental domain.

18.9 Implication for future classification

The phase-family classification problem can therefore be organized into three exact layers:

- (i) arithmetic classification of the rotation number as a function of (r, β) ;
- (ii) symbolic classification of the finite words realizing a given rational rotation number;
- (iii) geometric reconstruction of the corresponding complex periodic orbit and its stability.

The present manuscript solves the third step whenever an admissible word is specified and gives a finite exact procedure for the second step. The first step connects the problem to established contracted-rotation arithmetic and leaves a natural path toward a more complete global phase theorem.

19 Further structural consequences and scope

The exact formulas above suggest several structural principles that are worth isolating. They also clarify what is special about the monomial model and what should be regarded as a genuine open direction.

19.1 Why the monomial case is exactly solvable

The inverse relation $w^r = -z$ has two special features. First, all inverse images of a nonzero point have the same modulus. Second, their arguments form an exact arithmetic progression. In logarithmic polar coordinates

$$u = \log |z|, \quad \phi \in \mathbb{R},$$

a branch k acts by the affine map

$$(u, \phi) \mapsto \left(\frac{u}{r}, \frac{\phi + \pi + 2\pi k}{r} \right). \quad (137)$$

Thus the entire inverse tree is generated by a finite affine IFS. The regular polygons, base- r coding, radial convergence, and Bernoulli branch measures are not unrelated coincidences; they are consequences of this common affine structure.

Table 3: Structural features of the monomial inverse model.

Property	$-z^r$	az^r	General polynomial
Common inverse modulus	yes	yes	generally no
Regular-polygon inverse sets	yes	yes	generally no
Exact base- r coding	yes	yes	generally no
Finite-depth Fourier cancellation	yes	yes	generally no
Affine principal angular law	yes	phase-shifted	generally no
Uniform Bernoulli branch model	yes	yes	generally no

This comparison suggests a natural perturbative research program: identify which features survive under affine or nonlinear perturbations and which are rigid signatures of the monomial structure.

19.2 Piecewise-contraction and contracted-rotation context

The phase family should be viewed as a special, explicitly parameterized piecewise contraction rather than as a generic chaotic system. The broader literature contains examples where discontinuous contractions admit periodic attractors as well as exceptional Cantor dynamics [20, 19]. Contracted rotations provide an especially close comparison because they have a single affine slope and a modular jump [17].

The important logical point is that a general theorem from piecewise-contraction theory does not automatically become a theorem about the present root-induced family. The special family should instead be analyzed using its exact formulas, with general theory used for positioning and for identifying possible phenomena worth testing.

In the contracted-rotation normalization, the irrational-rotation-number alternative is particularly important. It provides a mathematically precise reason to leave open the possibility

of nonperiodic attractors for some phases, rather than declaring all branch-switching behavior chaotic or all parameters periodic. Whether the special affine relation

$$\delta_\beta = \frac{r-1}{2r} + \frac{\beta}{2\pi}$$

realizes every phenomenon available in the general contracted-rotation family is a separate question.

19.3 Parameter-space problems

For each r and p , let $\mathcal{W}_{r,p}$ denote the union of strict parameter windows associated with primitive chronological words of least period p . The exact window theorem gives several concrete problems:

- (a) determine all connected components of $\mathcal{W}_{r,p}$;
- (b) describe their ordering along the β axis;
- (c) determine when different symbolic representatives encode the same geometric orbit through cyclic shifts;
- (d) locate accumulation points of window endpoints;
- (e) classify the parameter values lying exactly on branch boundaries.

Because every endpoint is defined by an explicit affine equality, these questions are algebraic-combinatorial rather than purely numerical.

19.4 Entropy and nonperiodic regimes

For a parameter value one may define the finite-word complexity

$$p(n) := \#\{\text{admissible branch words of length } n\} \quad (138)$$

and the associated exponential complexity rate

$$h_{\text{top}} := \limsup_{n \rightarrow \infty} \frac{1}{n} \log p(n). \quad (139)$$

In the negative-root case the absolute-angle contraction forces a single asymptotic itinerary, so the corresponding asymptotic symbolic complexity is zero.

It is tempting to ask whether some phase produces positive topological entropy. The general piecewise-contraction literature shows that positive-entropy examples can exist in broader classes, but the present contracted-rotation subfamily is much more constrained. This paper therefore treats positive entropy here as a question requiring proof, not as an unproved feature of the model.

At the same time, the contracted-rotation literature shows that exceptional Cantor attractors are a legitimate possibility in the general setting when the associated rotation number is irrational [17]. Determining whether such an attractor can occur for the precise root-induced parameterization is an interesting and sharply formulated direction for future work.

19.5 A hierarchy of limiting viewpoints

The results admit a useful hierarchy:

$$\text{fixed-point problem} \subset \text{principal-orbit problem} \subset \text{phase-family problem}, \quad (140)$$

while the fully branched construction runs in parallel through inverse iteration and IFS geometry. The fixed negative phase is simple only after exploiting a special absolute-value symmetry; the phase family restores the branch-symbolic degrees of freedom that the symmetry removes.

This hierarchy helps prevent two common overstatements. First, the existence of a general contracted-rotation framework does not by itself prove that a particular root-induced phase realizes an arbitrary rotation number. Second, the existence of Cantor attractors in the broader class does not imply their existence in the present special family. Exact claims in this manuscript are always tied to the formulas actually proved for the root maps.

19.6 Originality and relationship with earlier complex-radical work

The subject of complex iterated radicals predates the present manuscript by many decades. Ogilvy, Gerber, Rohde, Walker, Jones, and Schuske–Thron provide classical treatments of convergence and periodic infinite radicals [2, 3, 4, 5, 6, 7]. Consequently, the contribution should not be described as the discovery of complex iterated radicals or of the basic negative-root two-cycle itself.

The mathematical contribution developed here is better characterized as a unified exact treatment of several layers that are often discussed separately: principal-branch dynamics, exact symbolic branch coding, inverse-tree geometry, weighted branch measures, sharp geometric asymptotics, variable orders, and the phase-parameter extension. The central originality claim should therefore remain deliberately conservative and should be evaluated against the relevant literature for each of these extensions separately.

19.7 Methodological lessons

Several reusable lessons emerge from the analysis:

- (i) a complex root is multivalued until a branch is specified;
- (ii) principal-value reduction must be performed at every iterate;
- (iii) an unwrapped angular recurrence is a legitimate lift, but not the same map as the principal dynamics;
- (iv) symbolic history order must be stated explicitly, because reversing a word changes its association with a geometric endpoint even when cardinalities are unchanged;
- (v) equality in distribution should not be silently replaced by pathwise equality in probabilistic branch arguments;
- (vi) density of finite branch sets does not imply pointwise convergence of one deterministic branch path;
- (vii) endpoint conventions matter exactly at discontinuity collisions;
- (viii) exact solvability in the monomial model comes from affine logarithmic coordinates rather than from a new arithmetic operation.

20 Common logical pitfalls and methodological lessons

20.1 A complex square root is not globally single-valued

Before a branch is selected, the equation $w^2 = z$ has two solutions for $z \neq 0$. Any iteration that uses “ $\sqrt{z}$ ” must specify which determination is intended.

20.2 The principal logarithm and the multivalued logarithm answer different questions

The notation

$$z^{1/r} = \exp((1/r) \log z)$$

with a multivalued logarithm describes all branch values, while

$$\text{Root}_r^{\text{pr}}(z) = \exp((1/r) \text{Log } z)$$

describes one principal determination.

20.3 Argument reduction must occur at every step

An affine recurrence for an unwrapped angle is a lifted model. It is not automatically the same as the principal argument dynamics. The negative-root case demonstrates this in the sharpest possible way.

20.4 Density is not pointwise convergence

The fact that A_n becomes dense in the unit circle does not imply that one chosen branch path converges through every point of the circle. A deterministic orbit, a finite branch set, and a branch measure are different mathematical objects.

20.5 Recursive labels do not create a new field

Writing i_n in place of a long radical does not create a new algebraic primitive. The genuinely additional structure lies in the branch history and in the dynamical rule used to generate it.

21 Worked exact examples and dependence on the root order

The closed formulas allow several representative orders to be compared without numerical iteration. This section records the resulting cycles and rates in a uniform notation and makes explicit how the order parameter controls the geometry.

21.1 The orders $r = 2, 3, 4, 5$

For the negative-root map, the attracting cycle is always

$$\mathcal{C}_r = \left\{ e^{i\pi/(r+1)}, e^{-i\pi/(r+1)} \right\}.$$

Thus the first few cases are Here the chord length is the Euclidean distance between the two

Table 4: Exact attracting cycles for several root orders.

r	cycle angles	chord length	two-step multiplier
2	$\pm\pi/3$	$\sqrt{3}$	$1/4$
3	$\pm\pi/4$	$\sqrt{2 - \sqrt{2}}$	$1/9$
4	$\pm\pi/5$	$2 \sin(\pi/10)$	$1/16$
5	$\pm\pi/6$	1	$1/25$

cycle points.

For example, when $r = 3$,

$$a_3 = e^{i\pi/4}, \quad b_3 = e^{-i\pi/4},$$

and

$$F_3(a_3) = b_3, \quad F_3(b_3) = a_3.$$

The magnitude recurrence is

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{3},$$

so

$$\alpha_n = \frac{\pi}{4} + \left(-\frac{1}{3}\right)^n \left(\alpha_0 - \frac{\pi}{4}\right).$$

Every nonzero orbit therefore approaches the same pair at the exact angular rate 3^{-n} .

21.2 A general initial-angle family

Suppose $z_0 = \rho_0 e^{i\theta_0}$ with $\theta_0 \neq \pi$. The sign alternates from the first step and

$$\theta_n = (-1)^n \left[\frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(|\theta_0| - \frac{\pi}{r+1}\right) \right] \quad (141)$$

when $\theta_0 > 0$, with the corresponding reflected formula for $\theta_0 < 0$. The modulus is simultaneously

$$\rho_n = \exp\left(\frac{\log \rho_0}{r^n}\right).$$

Hence the complete principal orbit is explicitly determined by the pair of scalar initial data $(\log \rho_0, |\theta_0|)$.

This separation has a useful interpretation. The initial modulus and initial angular magnitude contribute independently at the leading r^{-n} scale, and their Euclidean effects combine in the asymptotic coefficient from Theorem 81.

21.3 The square-root orbit from $z_0 = i$

For $r = 2$ and $\theta_0 = \pi/2$,

$$\alpha_* = \frac{\pi}{3}, \quad \alpha_0 - \alpha_* = \frac{\pi}{6}.$$

Since the initial angle is positive,

$$\theta_n = (-1)^n \frac{\pi}{3} + \frac{\pi}{6 \cdot 2^n}. \quad (142)$$

The first terms are

$$\frac{\pi}{2}, -\frac{\pi}{4}, \frac{3\pi}{8}, -\frac{5\pi}{16}, \frac{11\pi}{32}, \dots$$

and the exact complex orbit is

$$z_n = \exp\left(i\left((-1)^n \frac{\pi}{3} + \frac{\pi}{6 \cdot 2^n}\right)\right).$$

The orbit therefore alternates across the real axis while its angular displacement from the corresponding cycle point decays by exactly a factor 1/2 at every step.

21.4 Large-order behavior of the principal cycle

The order parameter also controls the location and strength of the attracting cycle. Since

$$\frac{\pi}{r+1} = \frac{\pi}{r} + O(r^{-2}),$$

the two cycle points approach 1 as $r \rightarrow \infty$. Their chord length satisfies

$$|a_r - b_r| = 2 \sin\left(\frac{\pi}{r+1}\right) = \frac{2\pi}{r+1} + O(r^{-3}). \quad (143)$$

Thus the two-cycle collapses toward the positive real axis at order r^{-1} .

At the same time the two-step multiplier is

$$r^{-2},$$

so the local attraction strengthens quadratically in the order. The order parameter therefore has two simultaneous effects: it moves the cycle toward 1 and accelerates attraction to that cycle.

Corollary 21.1 (Large-order cycle limit). *As $r \rightarrow \infty$,*

$$a_r \rightarrow 1, \quad b_r \rightarrow 1, \quad |a_r - b_r| \sim \frac{2\pi}{r}, \quad F'_r(a_r)F'_r(b_r) = r^{-2} \rightarrow 0. \quad (144)$$

21.5 Large-order behavior of the phase family

For the fixed-point regime of the phase family, the attracting angle is

$$\theta_*(\beta) = \frac{\beta}{r-1}.$$

For fixed β , this satisfies

$$\theta_*(\beta) = \frac{\beta}{r} + O(r^{-2}) \quad (145)$$

as $r \rightarrow \infty$. Hence the entire fixed-point branch collapses toward 1 in the complex plane as the root order grows.

The exact contraction ratio of the affine fixed-point dynamics is $1/r$, while the negative phase has two-step multiplier $1/r^2$. This provides a useful quantitative contrast between the ordinary fixed-point regime and the symmetric branch-switching regime.

21.6 An order-uniform synchronization estimate

For the negative-root system, every two principal orbits driven by the same root order r satisfy

$$|\alpha_n - \tilde{\alpha}_n| = r^{-n} |\alpha_0 - \tilde{\alpha}_0|. \quad (146)$$

Since $|\alpha_0 - \tilde{\alpha}_0| \leq \pi$, one obtains the uniform bound

$$|\alpha_n - \tilde{\alpha}_n| \leq \frac{\pi}{r^n}. \quad (147)$$

Thus the angular system has a global synchronization time scale set exactly by r^{-n} . The estimate is sharp at the scalar level because the slope of the absolute-angle map is exactly $-1/r$ everywhere.

Together with the radial identity,

$$|\log \rho_n - \log \tilde{\rho}_n| = \frac{1}{r^n} |\log \rho_0 - \log \tilde{\rho}_0|,$$

this shows that the full principal dynamics are globally contracting in logarithmic polar coordinates whenever one compares orbits using the same branch itinerary after the single exceptional zero-angle event.

21.7 Why the contraction does not imply point convergence

The scalar absolute-angle dynamics contract to a fixed point, yet the original signed complex orbit does not converge. The reason is that the scalar variable has forgotten the sign. The sign is restored by the deterministic branch rule, which reverses it at every nonzero step.

This gives a simple model example of a general principle: contraction of a quotient variable does not necessarily imply point convergence in the original state space. Here the quotient variable is $|\theta|$, the hidden discrete state is the sign, and the combined state converges to a two-cycle.

22 Synthesis of the main results

The preceding analysis can be summarized in one structural diagram. The principal map begins with a complex-valued operation but separates exactly into a radial contraction in logarithmic modulus and a discontinuous angular map. For the negative phase, the angular map admits a scalar absolute-value reduction and hence a unique nonzero two-cycle. Retaining all branches replaces the single-valued map by the inverse tree of a monomial, and logarithmic polar coordinates convert that inverse tree into an affine IFS. The same affine structure then reappears as base- r symbolic coding and as Bernoulli self-similar measure theory.

$$\begin{array}{c}
\text{principal root with branch convention} \\
\downarrow \\
\text{piecewise-affine angular contraction} \\
\downarrow \\
\text{exact periodic / phase structure} \\
\parallel \\
\text{symbolic branch coding of the inverse tree} \\
\downarrow \\
\text{regular polygons + Hausdorff limits + Bernoulli measures}
\end{array} \tag{148}$$

The fixed negative phase is therefore best understood not as an isolated curiosity but as the most rigid member of a larger parameterized architecture. At the same time, the fully branched model is not merely a numerical enlargement of the principal orbit: it is a different dynamical object with a different limiting geometry and, under probabilistic weighting, a different limiting measure.

23 Conclusion

The recursion

$$z_{n+1} = \text{Root}_r^{\text{pr}}(-z_n)$$

is an elementary map with a surprisingly rigid exact structure. The principal branch produces a discontinuity along the positive real axis, but polar coordinates separate the system into an exact radial law and a one-dimensional angular contraction. The radius satisfies

$$|z_n| = |z_0|^{1/r^n} \rightarrow 1,$$

while the angular magnitude satisfies

$$|\text{Arg } z_n| - \frac{\pi}{r+1} = \left(-\frac{1}{r}\right)^n \left(|\text{Arg } z_0| - \frac{\pi}{r+1}\right).$$

Consequently every nonzero principal orbit approaches the two-cycle

$$\left\{e^{i\pi/(r+1)}, e^{-i\pi/(r+1)}\right\}.$$

The cycle is locally holomorphically attracting with multiplier r^{-2} , and it is the unique nonzero periodic orbit.

The phase family

$$F_{r,\beta}(z) = \text{Root}_r^{\text{pr}}(e^{i\beta}z)$$

reveals that this two-cycle is a special consequence of the negative phase rather than a universal law of principal-root iteration. The periodic-itinerary formula turns branch reduction into a finite symbolic problem and permits explicit higher-period cycles, such as the period-three orbit for $r = 2$, $\beta = 2\pi/3$.

The fully multivalued recursion gives a different object: an exact inverse tree of the monomial $-z^r$. At depth n its endpoint set is a regular r^n -gon, converging to $\mathbb{S}^1$ in the Hausdorff metric. Uniform branch measures converge to Haar measure, while biased branch probabilities produce

non-Haar Bernoulli pushforwards whose entropy records information growth in the branch tree.

The distinction among fixed-point, pointwise, set-valued, and measure-valued limits is therefore not semantic decoration. It is the correct mathematical resolution of the phrase “infinite complex radical.”

The broader lesson is equally simple: branch conventions can be dynamical data. Once that viewpoint is adopted, a recursively generated complex radical can be studied simultaneously through ordinary complex analysis, symbolic coding, inverse iteration, geometric approximation, probability, and asymptotic dynamics—without enlarging the underlying complex field.

Final perspective. The interesting object is not a mysterious new number at infinite depth. It is the structured process that generates finitely many ordinary complex numbers, together with the orbit, tree, geometry, and measure induced by the branch rule.

A Appendix A: explicit formulas for signed angles

Suppose $\theta_0 > 0$ and $\theta_0 \neq \pi$. Then sign alternation gives

$$\theta_n = (-1)^n \alpha_n. \quad (149)$$

Substitution of (25) yields

$$\theta_n = (-1)^n \left[\frac{\pi}{r+1} + \left(-\frac{1}{r} \right)^n \left(\theta_0 - \frac{\pi}{r+1} \right) \right]. \quad (150)$$

If $\theta_0 < 0$, symmetry gives the corresponding formula after multiplication by -1 .

For $r = 2$, $\theta_0 = \pi/2$ gives

$$\theta_n = (-1)^n \frac{\pi}{3} + \frac{\pi}{6 \cdot 2^n},$$

as used in Section 8.

B Appendix B: exact Hausdorff calculation

Let

$$A = \{Re^{i(\phi+2\pi k/N)} : 0 \leq k < N\}.$$

Every $a \in A$ satisfies

$$\text{dist}(a, \mathbb{S}^1) = |R - 1|,$$

so

$$\sup_{a \in A} \text{dist}(a, \mathbb{S}^1) = |R - 1|.$$

For $u = e^{it} \in \mathbb{S}^1$, choose a nearest angular vertex. Then the angular difference is at most π/N , and

$$|u - a_k|^2 = 1 + R^2 - 2R \cos \delta, \quad |\delta| \leq \frac{\pi}{N}.$$

The cosine is decreasing on $[0, \pi]$, so the largest nearest-vertex distance occurs at $|\delta| = \pi/N$. Hence

$$\sup_{u \in \mathbb{S}^1} \text{dist}(u, A) = \sqrt{1 + R^2 - 2R \cos(\pi/N)}.$$

Taking the maximum gives (78).

C Appendix C: Fourier proof of equidistribution

Let $N = r^n$ and write the normalized unit-radius polygon as

$$\tilde{A}_n = \left\{ e^{i(\phi_n + 2\pi k/N)} : 0 \leq k < N \right\}.$$

For an integer ℓ ,

$$\frac{1}{N} \sum_{k=0}^{N-1} e^{i\ell(\phi_n + 2\pi k/N)} = e^{i\ell\phi_n} \frac{1}{N} \sum_{k=0}^{N-1} q^k, \quad q = e^{2\pi i\ell/N}.$$

If $N \nmid \ell$, then $q \neq 1$ while $q^N = 1$, so

$$\sum_{k=0}^{N-1} q^k = \frac{1 - q^N}{1 - q} = 0.$$

Thus every fixed nonzero Fourier coefficient vanishes for all sufficiently large n . Uniform density of trigonometric polynomials in $C(\mathbb{S}^1)$ gives weak convergence to Haar measure. The radial factor $R_n \rightarrow 1$ uniformly completes the proof in $\mathbb{C}$.

D Appendix D: proof of the variable-order formula

Starting from

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r_n},$$

one obtains by induction

$$\begin{aligned} \alpha_1 &= -\frac{\alpha_0}{r_1} + \frac{\pi}{r_1}, \\ \alpha_2 &= \frac{\alpha_0}{r_1 r_2} - \frac{\pi}{r_1 r_2} + \frac{\pi}{r_2}, \end{aligned}$$

and in general

$$\alpha_n = \frac{(-1)^n \alpha_0}{r_1 \cdots r_n} + \pi \left(\frac{(-1)^{n-1}}{r_1 \cdots r_n} + \frac{(-1)^{n-2}}{r_2 \cdots r_n} + \cdots + \frac{1}{r_n} \right),$$

which is exactly (113).

E Appendix E: principal phase family and branch words

For a prescribed word $(k_0, \dots, k_{p-1})$ the branch equations are

$$\theta_{j+1} = \frac{\theta_j + \beta - 2\pi k_j}{r}.$$

Repeated substitution gives

$$\theta_p = \frac{\theta_0}{r^p} + \frac{\beta}{r^p} (1 + r + \cdots + r^{p-1}) - \frac{2\pi}{r^p} (k_0 r^{p-1} + \cdots + k_{p-1}).$$

The periodic condition $\theta_p = \theta_0$ therefore yields

$$\theta_0 = \frac{\beta(1 + r + \cdots + r^{p-1}) - 2\pi(k_0 r^{p-1} + \cdots + k_{p-1})}{r^p - 1}.$$

The remaining problem is purely admissibility: at each stage one checks whether the corresponding branch index really performs the principal reduction. This finite collection of inequalities is the exact symbolic test.

F Appendix F: bibliographic and notational notes

F.1 Principal branch convention

The convention throughout is $\text{Arg } z \in (-\pi, \pi]$. Different software systems sometimes encode the negative-real-axis endpoint slightly differently, but this only changes a boundary representative. The off-boundary dynamics and all generic asymptotic statements are unchanged.

F.2 Terminology

The expressions “principal orbit,” “branch history,” “full branch set,” and “branch measure” are used as formal terms in this paper. They refer, respectively, to:

- (a) the single-valued orbit under F_r ;
- (b) the finite sequence of root choices;
- (c) the set of all endpoints after a fixed depth;
- (d) a probability-weighted distribution on those endpoints.

No new field operation is introduced.

G Appendix G: Detailed phase-window calculations

This appendix records the algebra underlying the periodic-window theorem in a form that can be used for exact verification of individual symbolic words. The point is not to replace the conceptual development in Section 7, but to make the finite calculations completely transparent.

G.1 Affine dependence on the phase parameter

Fix $r \geq 2$ and a word $\mathbf{k} = (k_0, \dots, k_{p-1})$. Write $b = \beta/\pi$ and $u_j = \theta_j/\pi$. The branch equations become

$$u_{j+1} = \frac{u_j + b - 2k_j}{r}. \quad (151)$$

Starting from an arbitrary affine expression $u_0(b) = A_0 b + B_0$, induction gives

$$u_j(b) = A_j b + B_j, \quad A_{j+1} = \frac{A_j + 1}{r}, \quad B_{j+1} = \frac{B_j - 2k_j}{r}. \quad (152)$$

Thus every candidate orbit coordinate depends affinely on b .

For the periodic candidate one has

$$u_0(b) = \frac{bS_p - 2K_p^{\text{chron}}}{r^p - 1}, \quad (153)$$

where $S_p = (r^p - 1)/(r - 1)$ and

$$K_p^{\text{chron}} = k_0 + k_1r + \cdots + k_{p-1}r^{p-1}. \quad (154)$$

The recurrence then determines $u_1, \dots, u_{p-1}$ uniquely.

G.2 The admissibility inequalities as an interval intersection

At step j , strict principal admissibility is

$$-1 < u_j(b) + b - 2k_j < 1. \quad (155)$$

Since the middle expression is affine in b , there are three possibilities: the inequality is impossible, it is automatically satisfied on the chosen parameter domain, or it defines an open interval.

Therefore the strict parameter set is a finite intersection of open intervals. Hence it is itself an open interval or the empty set. In particular, a fixed symbolic word cannot produce two disconnected strict windows.

Proposition G.1 (Exact endpoint mechanism). *Let $I_{\mathbf{k}}$ be a nonempty strict window lying in the interior of $(-\pi, \pi)$. Then each endpoint of $I_{\mathbf{k}}$ is obtained by solving*

$$u_j(b) + b - 2k_j = 1 \quad \text{or} \quad u_j(b) + b - 2k_j = -1 \quad (156)$$

for at least one $j \in \{0, \dots, p-1\}$.

Proof. If an endpoint failed all boundary equalities, every strict inequality would remain strict in a sufficiently small neighborhood, contradicting maximality of the interval. Conversely, every such equality is a candidate boundary; the complete window is obtained by intersecting all inequalities. $\square$

G.3 Cyclic representatives and the meaning of a word

A geometric period- p orbit admits p cyclic representations as a chronological word, one for each choice of starting point. These words have the same least period and generate the same set of orbit points, but generally produce different formulas for the distinguished starting angle θ_0 .

This explains an apparent asymmetry that can arise in symbolic calculations: two formulas associated with cyclic shifts need not agree term by term, even though their resulting geometric cycles agree after a cyclic relabeling. For this reason the paper fixes a chronological convention and never identifies cyclic words until the corresponding orbit has been constructed.

G.4 The period-three calculation in full

For $r = 2$ and $\mathbf{k} = (1, 0, 0)$,

$$S_3 = 7, \quad K_3^{\text{chron}} = 1,$$

so

$$u_0(b) = b - \frac{2}{7}. \quad (157)$$

The successive coordinates are

$$u_1(b) = \frac{u_0 + b - 2}{2} = b - \frac{8}{7}, \quad (158)$$

$$u_2(b) = \frac{u_1 + b}{2} = b - \frac{4}{7}. \quad (159)$$

The branch expressions appearing before division by 2 are

$$2u_0 + b - 2 = 2b - \frac{16}{7}, \quad u_1 + b = 2b - \frac{8}{7}, \quad u_2 + b = 2b - \frac{4}{7}.$$

Strict admissibility therefore requires

$$\begin{aligned} -1 &< 2b - \frac{16}{7} < 1, \\ -1 &< 2b - \frac{8}{7} < 1, \\ -1 &< 2b - \frac{4}{7} < 1. \end{aligned}$$

The corresponding intervals are

$$\frac{9}{14} < b < \frac{23}{14}, \quad \frac{1}{14} < b < \frac{15}{14}, \quad -\frac{3}{14} < b < \frac{11}{14}.$$

Their intersection is exactly

$$\frac{9}{14} < b < \frac{11}{14}. \quad (160)$$

At $b = 2/3$,

$$(u_0, u_1, u_2) = \left(\frac{8}{21}, -\frac{10}{21}, \frac{2}{21} \right),$$

which gives the period-three orbit in the main text.

G.5 Why strict admissibility gives local persistence

Let b_* lie in the interior of a strict window. By definition there is a positive margin ε such that every branch expression satisfies

$$-1 + \varepsilon \leq u_j(b_*) + b_* - 2k_j \leq 1 - \varepsilon.$$

Every $u_j(b)$ is affine in b , hence Lipschitz. Consequently there exists $\delta > 0$ such that $|b - b_*| < \delta$ preserves all inequalities. Thus the itinerary is locally constant and the periodic point varies affinely with the parameter.

This is the exact algebraic analogue of symbolic structural stability away from a discontinuity.

G.6 A rationality consequence

Suppose $\beta/\pi \in \mathbb{Q}$. Then δ_β in (46) is rational, while $\lambda = 1/r$ is rational. The normalized phase family is therefore a contracted rotation with rational parameters. The general contracted-rotation theory implies that its rotation number is rational, and in the rational-rotation case

the limit set is periodic [17]. Thus rational phase parameters belong to a particularly rigid arithmetic subfamily.

This statement is intentionally separated from the exact finite-word window theorem: the former uses established general theory, while the latter is proved directly from the branch equations of the present root system.

G.7 Algebraic parameters and the broader arithmetic picture

More generally, if β/π is algebraic, then δ_β is algebraic. Laurent and Nogueira prove that when both the contraction ratio and offset of a contracted rotation are algebraic, the associated rotation number is rational [17]. This places the phase family at an interesting boundary between exact algebraic dynamics and transcendental parameter dependence.

The conclusion should not be overstated: rational rotation number of the normalized system does not by itself provide the explicit symbolic word or the exact root-orbit formula. What this paper adds is the ability to compute such candidates directly from the original complex-root recurrence.

H Appendix H: Detailed branch coding and self-similar measures

H.1 Exact endpoint coding

Let $\mathcal{H}_n = \{0, \dots, r-1\}^n$ and let $\mathbf{k} = (k_1, \dots, k_n)$ be chronological. Repeated substitution in the inverse recurrence gives

$$\phi_n = \frac{\phi_0 + \pi S_n + 2\pi K_n^{\text{chron}}}{r^n}, \quad K_n^{\text{chron}} = k_1 + k_2 r + \dots + k_n r^{n-1}. \quad (161)$$

The digit map is bijective from $\mathcal{H}_n$ to $\{0, \dots, r^n - 1\}$. Consequently the symbolic tree and geometric polygon have exactly the same finite cardinality.

H.2 The reversal phenomenon in probability

The weighted endpoint angle can be written

$$\phi_n = \frac{\phi_0}{r^n} + \frac{\pi S_n}{r^n} + 2\pi \sum_{j=1}^n K_j r^{-(n-j+1)}. \quad (162)$$

The last sum is reversed relative to the usual forward base- r expansion. If the K_j are i.i.d., reversal leaves their joint law unchanged, so for every fixed n ,

$$\sum_{j=1}^n K_j r^{-(n-j+1)} \stackrel{d}{=} \sum_{j=1}^n K_j r^{-j}. \quad (163)$$

This identity is exact in distribution and is the precise reason the limiting weighted measure has the simple forward expansion description.

For non-i.i.d. digits this argument generally fails. The limiting object can then retain genuine dependence on the chronological ordering of the branch probabilities. Thus independence and stationarity are not superficial assumptions in the weighted-measure theorem.

H.3 Cylinder masses

For a word $\omega = (\omega_1, \dots, \omega_n)$, the corresponding base- r cylinder is

$$I_\omega = \left[\sum_{j=1}^n \frac{\omega_j}{r^j}, \sum_{j=1}^n \frac{\omega_j}{r^j} + \frac{1}{r^n} \right]. \quad (164)$$

Its probability is

$$\eta_p(I_\omega) = p_{\omega_1} \cdots p_{\omega_n}, \quad (165)$$

up to the usual ambiguity at endpoints with two base- r expansions. The cylinder diameter is exactly r^{-n} .

H.4 Proof of exact dimensionality

Assume first that $p_k > 0$ for every digit. By the strong law of large numbers, for η_p -almost every symbolic sequence ω ,

$$\frac{1}{n} \sum_{j=1}^n \log p_{\omega_j} \longrightarrow \sum_{k=0}^{r-1} p_k \log p_k = -h(p). \quad (166)$$

Hence

$$\frac{\log \eta_p(I_{\omega|n})}{-n \log r} \longrightarrow \frac{h(p)}{\log r}. \quad (167)$$

The balls around a non-endpoint coded point are bounded above and below by a uniformly bounded number of adjacent cylinders at comparable levels. Therefore cylinders and metric balls have the same logarithmic scaling exponent, proving

$$\lim_{\varepsilon \downarrow 0} \frac{\log \eta_p(B(x, \varepsilon))}{\log \varepsilon} = \frac{h(p)}{\log r} \quad (168)$$

for almost every x .

If some $p_k = 0$, restrict to the support alphabet. The same argument applies on the corresponding symbolic subshift, and the entropy formula remains valid because the zero terms are interpreted as $0 \log 0 = 0$.

The circle map $\Psi(x) = e^{i(2\pi x + \pi/(r-1))}$ is Lipschitz and finite-to-one, with ambiguity confined to the standard endpoint set. Therefore the exact dimension is preserved under the pushforward and

$$\dim_H \nu_p = \frac{h(p)}{\log r}. \quad (169)$$

H.5 Frequency classes and combinatorial dimension

Let q be a probability vector and let $E(q)$ consist of sequences for which the empirical digit frequencies converge to q . For a large n , the number of length- n words with empirical frequency close to q is asymptotic on the exponential scale to

$$\exp(nh(q)), \quad (170)$$

by Stirling's formula applied to multinomial coefficients. Each such cylinder has diameter r^{-n} . Therefore the corresponding frequency set has Hausdorff dimension

$$\dim_H E(q) = \frac{h(q)}{\log r}. \quad (171)$$

This provides the geometric side of the multifractal picture independently of the branch probabilities.

For the measure ν_p , a q -typical sequence has logarithmic cylinder mass

$$\frac{1}{n} \log \eta_p(I_{\omega|n}) \longrightarrow \sum_k q_k \log p_k, \quad (172)$$

so the local dimension is the cross-entropy quotient

$$d_{\nu_p}(x) = \frac{-\sum_k q_k \log p_k}{\log r}. \quad (173)$$

When $q = p$, this reduces to $h(p)/\log r$.

H.6 Uniform branch measure and Fourier cancellation

When $p_k = 1/r$, each length- n cylinder has mass r^{-n} and the pushforward is uniform on the r^n polygon vertices. For a fixed nonzero Fourier mode ℓ , once $r^n > |\ell|$,

$$\sum_{k=0}^{r^n-1} e^{2\pi i \ell k / r^n} = 0. \quad (174)$$

Thus finite-depth cancellation is exact, not merely asymptotic. The only residual geometric error comes from the radial displacement $R_n - 1$, which tends to zero exponentially.

I Appendix I: Detailed asymptotic expansions

I.1 Expansion of the radial factor

Let $L = \log \rho_0$. Then

$$R_n = e^{L/r^n} = 1 + \frac{L}{r^n} + \frac{L^2}{2r^{2n}} + O(r^{-3n}). \quad (175)$$

Consequently

$$R_n - 1 = \frac{L}{r^n} + \frac{L^2}{2r^{2n}} + O(r^{-3n}). \quad (176)$$

The exact radial error therefore has a complete asymptotic expansion in powers of r^{-n} .

I.2 Expansion of the angular Hausdorff term

Put $t = \pi/r^n$. Then

$$D_n^2 := R_n^2 + 1 - 2R_n \cos t \quad (177)$$

$$= (R_n - 1)^2 + 2R_n(1 - \cos t). \quad (178)$$

Using

$$1 - \cos t = \frac{t^2}{2} - \frac{t^4}{24} + O(t^6)$$

and (175), one obtains

$$D_n^2 = \frac{L^2 + \pi^2}{r^{2n}} + \frac{L^3 + L\pi^2}{r^{3n}} + O(r^{-4n}). \quad (179)$$

Hence

$$D_n = \frac{\sqrt{L^2 + \pi^2}}{r^n} + \frac{L(L^2 + \pi^2)}{2(L^2 + \pi^2)^{1/2}} \frac{1}{r^{2n}} + O(r^{-3n}), \quad (180)$$

provided one interprets the expansion with the standard positive square-root branch. In particular the first-order constant is exactly $\sqrt{L^2 + \pi^2}$.

I.3 Dominance of the angular side

The radial one-sided Hausdorff term satisfies

$$|R_n - 1| = \frac{|L|}{r^n} + O(r^{-2n}), \quad (181)$$

whereas the angular term satisfies

$$D_n = \frac{\sqrt{L^2 + \pi^2}}{r^n} + o(r^{-n}). \quad (182)$$

Because $\sqrt{L^2 + \pi^2} > |L|$, the angular term is eventually the larger one. Thus the max in the exact Hausdorff formula eventually disappears and the Hausdorff distance itself has the sharp expansion.

I.4 Individual orbit expansion

Let

$$A := \left| |\theta_0| - \frac{\pi}{r+1} \right|, \quad L = \log \rho_0.$$

For the matched cycle point c_n ,

$$|\theta_n - \arg c_n| = Ar^{-n}. \quad (183)$$

Combining this with the radial expansion yields

$$|z_n - c_n|^2 = \frac{L^2 + A^2}{r^{2n}} + O(r^{-3n}), \quad (184)$$

so

$$|z_n - c_n| = \frac{\sqrt{L^2 + A^2}}{r^n} + O(r^{-2n}). \quad (185)$$

This refines the global $O(r^{-n})$ statement to a complete leading-order asymptotic.

I.5 A special unit-modulus simplification

When $\rho_0 = 1$, one has $L = 0$. The Hausdorff constant reduces to π , and the individual-orbit constant reduces to the pure angular deviation A . The formulas in this special case explain why angular asymptotics alone can appear exact in unit-circle examples while becoming incomplete for general initial moduli.

J Appendix J: Detailed nonautonomous order analysis

J.1 Closed formula by repeated substitution

Starting with

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r_{n+1}}, \quad (186)$$

we obtain

$$\begin{aligned} \alpha_1 &= -\frac{\alpha_0}{r_1} + \frac{\pi}{r_1}, \\ \alpha_2 &= \frac{\alpha_0}{r_1 r_2} - \frac{\pi}{r_1 r_2} + \frac{\pi}{r_2}, \\ \alpha_3 &= -\frac{\alpha_0}{r_1 r_2 r_3} + \frac{\pi}{r_1 r_2 r_3} - \frac{\pi}{r_2 r_3} + \frac{\pi}{r_3}. \end{aligned}$$

The pattern gives

$$\alpha_n = \frac{(-1)^n \alpha_0}{R_n} + \pi \sum_{j=1}^n \frac{(-1)^{n-j}}{r_j r_{j+1} \cdots r_n}. \quad (187)$$

J.2 Unconditional forgetting

If α_n and $\tilde{\alpha}_n$ follow the same order sequence, their difference obeys

$$\alpha_{n+1} - \tilde{\alpha}_{n+1} = -\frac{1}{r_{n+1}}(\alpha_n - \tilde{\alpha}_n). \quad (188)$$

Therefore

$$\boxed{|\alpha_n - \tilde{\alpha}_n| = \frac{|\alpha_0 - \tilde{\alpha}_0|}{R_n} \leq 2^{-n} |\alpha_0 - \tilde{\alpha}_0|.} \quad (189)$$

No additional growth or regularity assumption on the order sequence is needed beyond $r_n \geq 2$.

J.3 Periodic schedules

If $r_{n+p} = r_n$, then the p -step affine map is

$$\alpha_{n+p} = a_p + \frac{(-1)^p}{r_1 \cdots r_p} \alpha_n \quad (190)$$

for an explicit forcing constant a_p . Its unique fixed point is the initial value of the unique periodic magnitude profile. The p -step error therefore satisfies

$$|\alpha_{n+p} - \alpha_0^*| \leq (r_1 \cdots r_p)^{-1} |\alpha_n - \alpha_0^*|. \quad (191)$$

Iterating over blocks of length p gives geometric convergence.

J.4 Mixed-order branch cardinalities

With $R_n = \prod_{j=1}^n r_j$ and

$$E_n = 1 + r_1 + r_1 r_2 + \cdots + r_1 \cdots r_{n-1},$$

the full branch set satisfies

$$A_n = \{w : w^{R_n} = (-1)^{E_n} z_0\}. \quad (192)$$

Since $z_0 \neq 0$, there are exactly R_n distinct endpoints. Thus the same cumulative product governs both the principal forgetting rate and the multibranch cardinality.

J.5 Periodic forcing and signed periods

When the magnitude profile has least period q and no profile value vanishes, the sign changes at each step. After q steps the magnitude returns to its original value; the sign returns precisely when q is even. If q is odd, one additional cycle is required. Consequently the signed period is q for even q and $2q$ for odd q .

This statement is sharper than the crude divisibility bound by $\text{lcm}(p, 2)$ because it uses the least period of the actual magnitude profile rather than merely the period of the forcing schedule.

J.6 Nonperiodic order schedules

For arbitrary order sequences the exact forcing term

$$B_n := \pi \sum_{j=1}^n \frac{(-1)^{n-j}}{r_j \cdots r_n} \quad (193)$$

is the complete source of nonautonomous complexity after the initial condition has been forgotten. Indeed,

$$\alpha_n - \frac{(-1)^n \alpha_0}{R_n} = B_n. \quad (194)$$

Since $R_n \rightarrow \infty$, two solutions with different initial angles become indistinguishable at the exponential scale 2^{-n} . Questions about asymptotic profiles for irregular order schedules therefore reduce to the study of the forcing sequence B_n .

References

- [1] P. B. Pulickal, “Stable 2-Cycles in the Iteration of the Principal Negative Complex Root: An Analysis of $z_{k+1} = (-z_k)^{1/n}$,” preliminary preprint, ResearchGate, October 2025. DOI: 10.6084/m9.figshare.30296524. https://www.researchgate.net/publication/396270980_Stable_2-Cycles_in_the_Iteration_of_the_Principal_Negative_Complex_Root_An_Analysis_of_z_k1_-z_k
- [2] C. S. Ogilvy, “To What Limits Do Complex Iterated Radicals Converge?” *The American Mathematical Monthly* 77 (1970), 388–389. DOI: 10.1080/00029890.1970.11992497.
- [3] L. Gerber, “Complex iterated radicals,” *Proceedings of the American Mathematical Society* 41 (1973), no. 1, 205–210. DOI: 10.1090/S0002-9939-1973-0318721-1.
- [4] H.-W. Rohde, “Complex Iterated Radicals,” *The American Mathematical Monthly* 81 (1974), no. 1, 14–21.
- [5] P. L. Walker, “Iterated complex radicals,” *The Mathematical Gazette* 67 (1983), no. 442, 269–273. DOI: 10.2307/3617262.

- [6] D. J. Jones, “Continued Powers and Roots,” *The Fibonacci Quarterly* 29 (1991), no. 1, 37–46. DOI: 10.1080/00150517.1991.12429454.
- [7] G. Schuske and W. J. Thron, “On periodic infinite radicals,” *Annales Fennici Mathematici*, no. 307 (1962), 1–8. DOI: 10.5186/aasfm.1962.307.
- [8] NIST Digital Library of Mathematical Functions, Chapter 4, “Elementary Functions,” §4.2, Logarithm, Exponential, Powers. <https://dlmf.nist.gov/4.2>.
- [9] S. G. Moreno and E. M. García-Caballero, “Chebyshev polynomials and nested square roots,” *Journal of Mathematical Analysis and Applications* 394 (2012), no. 1, 61–73. DOI: 10.1016/j.jmaa.2012.04.065.
- [10] N. Hemmingsson, “Equidistribution of iterations of holomorphic correspondences and Hutchinson invariant sets,” *Conformal Geometry and Dynamics* 28 (2024), 97–114. DOI: 10.1090/ecgd/394.
- [11] L. Pathirana, C. L. Petersen, and D. Vardakis, “Julia Sets for Sequences of Monomials,” arXiv:2607.23353 (2026). <https://arxiv.org/abs/2607.23353>.
- [12] A. F. Beardon, *Iteration of Rational Functions: Complex Analytic Dynamical Systems*, Graduate Texts in Mathematics 132, Springer, 1991.
- [13] J. W. Milnor, *Dynamics in One Complex Variable*, 3rd ed., Annals of Mathematics Studies 160, Princeton University Press, 2006.
- [14] K. J. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 2nd ed., Wiley, 2003.
- [15] P. Billingsley, “Hausdorff dimension in probability theory,” *Illinois Journal of Mathematics* 4 (1960), 187–209.
- [16] P. Billingsley, “Hausdorff dimension in probability theory II,” *Illinois Journal of Mathematics* 5 (1961), 291–298.
- [17] M. Laurent and A. Nogueira, “Rotation number of contracted rotations,” *Journal of Modern Dynamics* 12 (2018), 175–191. DOI: 10.3934/jmd.2018007.
- [18] A. Nogueira, B. Pires, and R. A. Rosales, “Topological dynamics of piecewise λ -affine maps,” *Ergodic Theory and Dynamical Systems* 38 (2018), no. 5, 1876–1893.
- [19] B. Pires, “Symbolic dynamics of piecewise contractions,” *Nonlinearity* 32 (2019), no. 12, 4871–4889. DOI: 10.1088/1361-6544/ab4361.
- [20] S. Jain and C. Liverani, “Piecewise contractions,” *Ergodic Theory and Dynamical Systems* 45 (2025), no. 5, 1503–1540. DOI: 10.1017/etds.2024.78.